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Looking Beyond the Government's Regulatory Toolkit

Early Findings on How Business and Civil Society Actors
Can Help Manage and Mitigate Transformative Artificial
Intelligence Risks and Impacts

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About This Report

Transformative artificial intelligence (AI)—that is, advanced systems that have the potential for broad societal impact—presents safety and societal risks, as well as prospective benefits. Although governments retain an indispensable role in governing AI, business and civil society organizations play material roles in how AI is designed, deployed, used, and monitored. For these reasons, in this report, we examine how business and civil society actors can help mitigate the societal and safety risks of transformative AI in ways that complement government action.

More concretely, we develop a structured framework outlining actions and responsibilities nongovernmental actors can take. Drawing on an existing body of literature, stakeholder interviews, and observations from RAND-designed games, we specify three overarching roles for nongovernmental actors: (1) managing technical and operational risks within development and deployment, (2) shaping incentives for safety through market and network mechanisms, and (3) supporting social stability and public trust during periods of AI-related change. Under each role, we detail potential responsibilities that include internal risk-management systems, incident reporting, safety-oriented procurement, workforce transition planning, and community resilience measures. We also identify actions that can inform or reinforce public policy and complement governmental roles.

This report is intended for business and civil society audiences seeking to understand how nongovernmental actors can contribute to mitigating the societal and safety risks of transformative AI and for researchers investigating governance roles beyond that of the state. We draw on and synthesize a substantial body of literature on AI safety, governance, and policy, and we have written it to be accessible to readers entering these fields without assuming prior expertise.

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Summary

In this report, we examine how business and civil society actors can help mitigate the societal and safety risks of transformative artificial intelligence (AI) in ways that complement, rather than replace, government action. We integrate a targeted literature review with semistructured interviews and observations from RAND-designed games to identify roles that are actionable under existing legal and market conditions, the enabling “plumbing” that is required for safety, and levers that can raise the level of baseline practice ahead of regulation. Our analysis focuses on measures that could plausibly be implemented today and notes where government authority remains essential.

Our research finds that business and civil society can contribute to mitigating risks by assuming three overarching roles, some of which have already been adopted by actors and institutions in the ecosystem:

- **Managing technical and operational risks within development and deployment.** Organizations that develop or deploy AI systems can establish systematic procedures to test systems prior to release, monitor them in operation, document safety cases and model inventories, and maintain clearly defined *break-glass* mechanisms for suspending high-risk systems when warranted.
- **Helping shape incentives for safety through market and network mechanisms.** Major purchasers, investors, insurers, and industry consortia can incorporate safety criteria into procurement, due diligence, and assurance processes, thereby rewarding developers that adopt robust risk-management practices and helping to diffuse common standards and norms for safe AI use across markets and sectors.
- **Supporting social stability and public trust during periods of AI-related change.** Employers, philanthropies, unions, professional bodies, and community organizations can plan for workforce transitions, invest in reskilling and redeployment, strengthen local resilience to AI-related shocks, and provide clear, credible communication about AI risks and benefits to sustain public confidence and social cohesion.

Table S.1 summarizes, at a high level, areas in which business and civil society can lead in mitigating and managing risks, and how those actions can support government.

TABLE S.1

Three Nongovernmental Roles in Governing Transformative AI with Interfaces to Government

Role	Purpose	Possible Actions for Business and Civil Society	Possible Government Interface
1. Manage technical and operational risks	Prevent foreseeable failures while models are built and deployed	• Board-level AI oversight • Rigorous pre- and postdeployment testing and monitoring • Confidential incident and hazard reporting • Break-glass rehearsals	• Enable evidence-driven oversight • Sandboxes and memorandums of understanding (controlled trials and access to frontier models) • Critical infrastructure alignment (data center resilience, redundancy, and zoning and permitting levers)
2. Shape business and market incentives	Use market power to make safety the winning strategy	• Procurement criteria and buyers' clubs • Investor and insurer diligence • Warranties • Third-party assurance and rating signals • Multistakeholder norms	• Public procurement as a force multiplier • Liability and insurance enablers • Competition and concentration oversight
3. Support social stability and public trust	Cushion social shocks and sustain legitimacy	• Workforce transition plans (reskilling, redeployment) • Community resilience partnerships • Crisis communications preparation • Philanthropy to support	• Co-fund and coordinate transitions • Create policy pathways for benefit sharing • Institute networks for cross-border coordination

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Looking Beyond the Government’s Toolkit

The rapid advance of artificial intelligence (AI) is reshaping expectations about future economic, political, and military power. Leading companies in the United States, China, and Europe are pursuing increasingly capable systems, and some are explicitly aiming at artificial general intelligence (AGI). Properly developed and governed, these systems could spur productivity, accelerate scientific discovery, and create tools to address enduring public-policy challenges. At the same time, these systems could introduce serious risks, from destabilizing military innovations and pervasive disinformation to labor disruption and social dislocation; in the most extreme scenarios, some analysts argue that they could generate catastrophic or even existential risks.

Many of the scientists and technologists who pioneered modern AI—including Nobel laureate Geoffrey Hinton¹—have warned that advanced AI could pose such risks. Prominent industry leaders have also publicly assigned nontrivial probabilities to extreme outcomes; for example, Anthropic chief executive officer Dario Amodei has suggested that the chance of catastrophic failure could be on the order of one in four.² Whether one accepts these specific estimates, the underlying policy question is straightforward: How should societies manage safety and societal risks from increasingly capable AI while preserving benefits?

Most research and public debate start from the premise that governments should lead in managing risk. Yet several dynamics now test the assumption that government will necessarily lead. Frontier research and development (R&D) and AI deployment are concentrated in private firms, technical progress is outpacing policy and regulation cycles,³ and the surfaces of impact lie largely outside direct state control. Given that frontier AI is being developed and deployed in contexts in which government is neither the sole nor the first mover, understanding the extent to which and how nongovernmental institutions, across the business community and civil society, can lead or support the management of safety and societal risks is an urgent research priority.

This report addresses that gap by specifying where nongovernmental institutions can lead, where they should support, and how they can complement government in ways that are feasible, effective, and democratically legitimate. We treat business and civil society as central actors in governing AI in a setting in which innovation is largely driven by industry. Methodologically, we synthesize the literature and incorporate original empirical evidence (from semistructured interviews and observations from RAND-designed games) to map roles and responsibilities and assess incentives and constraints within a practical framework.

For clarity, we use the following definitions:

- *AI* refers to computational systems that, for explicit or implicit human-defined objectives, infers from inputs how to generate outputs—such as predictions, content, recommendations, or decisions—that can

¹ Amanpour & Company, “Geoffrey Hinton Warns of the ‘Existential Threat’ of AI,” video, May 9, 2023.

² Megan Morrone, “Amodei on AI: ‘There’s a 25% Chance That Things Go Really, Really Badly,’” *Axios*, September 17, 2025.

³ State-level activity has been increasing: During the 2025 legislative session, lawmakers in all 50 states introduced 1,208 AI-related bills, the first year in which every state had introduced at least one, and 145 of those bills were enacted into law, including California’s Senate Bill 53 and New York’s RAISE Act, both of which require frontier-model developers to publish safety frameworks and report critical incidents. However, state-level measures vary widely in scope and stringency, federal policy remains unsettled, and the resulting patchwork leaves significant gaps that nongovernmental action can help fill. See “Artificial Intelligence (AI) Legislation Tracker 2026: All 50 States,” webpage, MultiState.ai, undated; Governor Gavin Newsom, “Governor Newsom Signs SB 53, Advancing California’s World-Leading Artificial Intelligence Industry,” news release, September 29, 2025; and Governor Kathy Hochul, “Governor Hochul Signs Nation-Leading Legislation to Require AI Frameworks for AI Frontier Models,” news release, December 19, 2025.

influence physical or virtual environments; such systems vary in their levels of autonomy and adaptiveness after deployment.⁴

- *Transformative AI* denotes advanced, general-purpose AI systems whose diffusion could plausibly drive large, systemic changes across critical sectors or social institutions within a relatively short time horizon.
- *AGI* commonly refers to highly capable systems that match or exceed human performance across most cognitive domains.⁵

Although the boundaries around these terms are contested, these working definitions anchor a policy-relevant discussion of risk and governance. We favor the term *transformative AI* over AGI because the former centers the conversation on the effects and consequences of developing and diffusing advanced AI systems rather than on empirically fraught and conceptually controversial debates about what constitutes general intelligence. Moreover, our analysis is inclusive of risks and potential harms, including labor disruptions, that arise from advanced systems that fall short of the AGI threshold.

Throughout this report, we refer to business and civil society actors and institutions. Although our investigation is not focused on the actions that each individual actor can take, our analysis is inclusive of the following categories:

- *Business actors* include firms that design and train advanced models; companies that integrate, fine-tune, or deploy AI in products and services; and the upstream and downstream commercial ecosystem that enables these activities. The category therefore spans frontier developers and platform providers; cloud and data-center operators and other critical-infrastructure firms; chip and tooling vendors; sectoral adopters in such domains as health care, finance, education, logistics, and media; and the market intermediaries that shape incentives and assurance, such as major buyers and procurers, insurers and reinsurers, credit-rating and audit firms, professional services and consultancies, and industry associations or consortia.
- *Civil society actors* comprise primarily nonstate and nonprofit organizations that articulate public values, generate independent expertise, or deliver services. This includes advocacy and watchdog organizations, research institutes and universities, professional bodies and standards-setting communities, philanthropic funders, journalist and investigative outlets, unions and worker organizations, community-based and faith-based groups, and open-source and volunteer technical communities. We treat these actors as analytically distinct from government while recognizing that many collaborate with public authorities, operate in coregulatory arrangements, or receive public funding.

Methods, Findings, and Observations

We used a mixed-methods design to triangulate insights and observations from the literature, stakeholder interviews, and RAND-designed games. First, we conducted a targeted review of scholarship to situate non-governmental governance mechanisms for transformative AI and inform the methodological approach to the interviews. We relied on academic and policy literature, focusing on publications about AI governance

⁴ Definitions of AI vary. This definition borrows from one introduced by the Organisation for Economic Co-operation and Development, “Recommendation of the Council on Artificial Intelligence,” OECD/LEGAL/0449, amended May 3, 2024.

⁵ There continues to be debate among researchers about how best to define AGI. See, for example, Dan Hendrycks, Dawn Song, Christian Szegedy, Honglak Lee, Yarin Gal, Erik Brynjolfsson, Sharon Li, Andy Zou, Lionel Levine, Bo Han, et al., “A Definition of AGI,” arXiv, arXiv:2510.18212, version 2, revised October 23, 2025.

and business management studies, as well as on relevant case studies in other domains. We identified relevant literature through iterative keyword searches across major academic databases, leading policy research outlets, and practitioner-focused publications. Search terms combined concepts related to *AI governance*, *transformative or frontier AI*, *private-sector and civil-society governance mechanisms*, *corporate responsibility*, *standards and assurance*, and *multistakeholder governance* and were refined over multiple rounds as key themes emerged. We prioritized literature published within the past two years to capture rapidly evolving debates, while also including earlier foundational works where necessary to establish conceptual baselines. Notably, the field of AI governance and safety moves quickly. Moreover, often excellent ideas are published in opinion pieces, blogs, and tweets.⁶ Because of both speed and the variety of non-peer-reviewed platforms on which leading ideas are transmitted, it can be difficult to give a full picture of what experts are saying in a report that privileges formal academic and policy publications. Therefore, this review should be seen as representative, not exhaustive, and as a selection of key literature for business and civil society actors newer to the field.

We then conducted 15 semistructured interviews with practitioners and actors from civil society and business selected through purposive sampling to capture variation across sectors and functions relevant to non-state governance of AI risks. Participants included business leaders and practitioners in health care, finance, and consulting who were involved in strategy, responsible AI, and product safety; civil-society leaders and researchers working in governance and higher education; former government officials with responsibilities spanning cybersecurity, export controls, and state-level policy; and frontier AI safety experts working in civil society and frontier AI labs. Participants were predominantly senior in their roles, including vice presidents, presidents and chief executive officers, company founders, former government leads, and senior product or policy managers. Appendix A provides more information on our methods, including demographic information, limitations to the approach, and interview protocols.

To observe decisionmaking under pressure, we also ran a series of tabletop simulation games that put participants in fast-moving AI crisis scenarios. In total, we facilitated five “Day After AGI” exercises (approximately two hours each) using RAND’s Infinite Potential gaming platform. Each game convened between ten and 14 senior leaders from industry and civil society to role-play an emergency response to a sudden AI-related crisis. Participants were chosen for their institutional experience and expertise (e.g., business executives, philanthropic and community leaders, former officials, and technical experts) to ensure informed discussion. The Business and Civil Society game simulated a special “Business and Civil Society Council” that was convened by the U.S. President in the wake of an AI-related crisis. In each case, the scenario presented a rapidly unfolding crisis triggered by advanced AI—an event posing immediate national or economic security threats and serving as a signpost toward a transformative AI future.

Participants, guided by a facilitator (playing a National Security Advisor role), were briefed on the scenario and tasked with diagnosing its implications, discussing possible responses and cross-sector actions, and ultimately advising national leadership on a path forward. Notably, the scenarios were set in the near-term future (between 12 and 18 months ahead) to heighten plausibility and urgency; aside from the AI shock described, geopolitical and technological conditions were assumed to remain largely as they were at the time of this writing in early 2026. Appendix B provides more information on the methodological approach. An after-action report providing full analysis of the five games has been published separately.⁷

⁶ Examples include writing by Helen Toner, Dean Ball, and Sebastien Krier on their Substacks and X accounts.

⁷ Richard S. Girven, Beba Cibralic, and Gregory Smith, *Infinite Potential—Insights from the Business and Civil Society Scenario: After-Action Report from a Sequence of Day After Artificial General Intelligence Exercises*, RAND Corporation, RR-A4767-1, 2026.

In the sections that follow, we summarize key insights from the literature and empirical findings that help inform the analytical framework presented in this report.

Summary of Key Literature

In this review, we survey leading work on nongovernmental approaches to AI risk management. We organize the literature using four categories that move from mechanisms closest to where AI systems originate to those that operate at progressively greater distance from the technology itself: (1) internal risk management and safety governance, (2) independent evaluation and assurance, (3) multistakeholder coordination and norm-setting, and (4) civil society oversight and public accountability. These categories capture a spectrum of governance functions—from the technical and organizational to the political and societal—that the literature identifies as necessary complements to government action.

Corporate Safety and Governance Practices

Frontier AI developers are increasingly expected to institute internal risk management practices to ensure that their technologies are safe and trustworthy.⁸ Best practices for frontier labs are increasingly framed as a continuous risk-management life cycle: predeployment and postdeployment evaluations, red-teaming and dangerous-capability testing, incident response and cybersecurity, independent or third-party assessment, meaningful usage controls, and transparency artifacts that allow external scrutiny.⁹

Leading frontier labs have published structured safety frameworks that formalize these practices.¹⁰ Anthropic’s Responsible Scaling Policy, introduced in 2023, was among the first to articulate a tiered approach in which models are assessed against defined capability levels, and advancing to higher tiers triggers increasingly stringent containment and deployment requirements. Google DeepMind’s Frontier Safety Framework follows a similar logic, identifying critical capability levels and specifying the mitigations that are required before deployment. OpenAI’s Preparedness Framework establishes internal risk scorecards across categories that include cybersecurity and biological threats, and there are thresholds that determine whether a model can proceed. These frameworks share a common structure—defined evaluation criteria, escalation pathways when risks are detected, and postdeployment monitoring—although how rigorously these frameworks are applied is difficult to assess from outside, and published safety reports vary considerably in specificity.

Adopting safety frameworks and design principles is one thing; sustaining them organizationally is another. The question is how firms institutionalize risk governance so that safety commitments survive pressure from competing priorities. We have already seen these safety frameworks at the labs change sub-

⁸ Public-opinion synthesis work by the Centre for the Governance of AI (GovAI) (which draws on a large body of survey evidence focused on the United States and United Kingdom) highlights that many people are more concerned than optimistic about AI’s societal impacts (including job disruption), support stronger oversight, and yet often distrust both tech companies and governments to manage AI well, particularly when acting alone. For more, see Noemi Dreksler, Harry Law, Chloe Ahn, Daniel S. Schiff, Kaylyn Jackson Schiff, and Zachary Peskowitz, *What Does the Public Think About AI? An Overview of the Public’s Attitudes Towards AI and a Resource for Future Research*, Centre for the Governance of AI, Future of Humanity Institute, University of Oxford, January 22, 2025.

⁹ Primary references on frontier-model safety governance include the AI Seoul Summit’s “Frontier AI Safety Commitments” (“Seoul Commitments”) and METR’s Frontier AI Safety Policies tracker, which indexes public responsible-scaling policies and frontier safety frameworks from leading developers. For a broader expert synthesis of advanced-AI risks and mitigations, see Yoshua Bengio, Stephen Clare, Carina Prunkl, Maksym Andriushchenko, Ben Bucknell, Malcolm Murray, Shalaleh Rismani, Conor McGlynn, Nestor Maslej, Philip Fox, et al., *International AI Safety Report 2026*, February 2026. For more, see Department for Science, Innovation and Technology, “Frontier AI Safety Commitments, AI Seoul Summit 2024,” United Kingdom government guidance, last updated February 7, 2025; METR, “Frontier AI Safety Policies,” webpage, undated.

¹⁰ Anthropic, “Anthropic’s Responsible Scaling Policy,” webpage, version 2.2, effective May 14, 2025; OpenAI, *Preparedness Framework*, version 2, last updated April 15, 2025; Google DeepMind, “Frontier Safety Framework,” version 3.1, April 17, 2026.

stantively. For example, on February 24, 2026, Anthropic released a substantially updated version 3.0 of its Responsible Scaling Policy. The third version is a meaningful departure: Anthropic argues that some commitments in the old responsible scaling policy make sense only if they are matched by other companies, and Anthropic also dropped its pause commitment.

Scholars have explored how AI firms can institutionalize safety practices within more-durable corporate structures. Jonas Schuett, for example, proposes applying the three lines of defense model—a widely used corporate risk management framework—to AI developers.¹¹ In this model, AI firms would assign roles for frontline risk controls, independent risk oversight functions, and internal audit, thereby closing gaps and enabling boards of directors to oversee AI risk management more effectively. The three-lines model was designed for financial institutions, where risks are actuarially tractable, and audit functions have established professional standards. Its application to AI development companies—where threat categories are poorly characterized and safety cases lack equivalent precedent—is not quite so cleanly described; organizations adopting this model for AI risk should account for this disanalogy when calibrating the scope and authority of each line. In particular, the second and third lines will need to develop evaluative standards that do not yet exist in mature form.¹²

Related studies examine the design of AI ethics boards within companies, identifying key questions around boards' mandates, composition, and authority, as well as the resources needed to reduce AI-related risks.¹³ The lessons from prior failed ethics boards indicate that such bodies must be empowered and well-structured to have any meaningful impact; absent real authority, they function as reputation management rather than risk management.¹⁴

On the technical side, making any of these governance structures operational requires the use of tools that do not yet fully exist. Anka Reuel and coauthors present a taxonomy of open problems in technical AI governance, including how to reliably watermark AI-generated content to detect deepfakes, continuously monitor deployed models for emergent harmful behaviors, and verify compliance with safety standards.¹⁵ Addressing these technical gaps will require dedicated R&D and cross-industry adoption of new methods, underscoring that the private sector and independent experts must invest in innovative safety mechanisms (such as robust model monitoring and content-provenance techniques) to support effective oversight of advanced AI systems.¹⁶

¹¹ Jonas Schuett, "Three Lines of Defense Against Risks from AI," *AI and Society*, Vol. 40, No. 2, February 2025.

¹² Noam Kolt and colleagues propose one answer to this gap: *legal alignment* or intentionally designing AI models to comply with applicable laws and fundamental rights as part of their safety architectures. This approach leverages the clarity of democratically legitimized laws but inherits their limitations, including jurisdictional variation and regulatory lag. See Noam Kolt, Nicholas Caputo, Jack Boeglin, Cullen O'Keefe, Rishi Bommasani, Stephen Casper, Mariano-Florentino Cuéllar, Noah Feldman, Iason Gabriel, Gillian K. Hadfield, et al., *Legal Alignment for Safe and Ethical AI*, Oxford Martin School, AI Governance Initiative, University of Oxford, January 15, 2026.

¹³ Jonas Schuett, Ann-Katrin Reuel, and Alexis Carlier, "How to Design an AI Ethics Board," *AI and Ethics*, Vol. 5, No. 2, April 2025.

¹⁴ Beba Cibralic, Miranda Bogen, Kevin Bankston, and Ruchika Joshi, *Principled Practice: A Playbook for Operationalizing Responsible AI*, Center for Democracy & Technology, AI Governance Lab, September 17, 2025.

¹⁵ Anka Reuel, Ben Bucknall, Stephen Casper, Tim Fist, Lisa Soder, Onni Aarne, Lewis Hammond, Lujain Ibrahim, Alan Chan, Peter Wills, et al., "Open Problems in Technical AI Governance," arXiv, arXiv:2407.14981, version 2, April 16, 2025.

¹⁶ An increasingly important set of open technical questions concerns AI agents. On the infrastructure needed to govern autonomous AI agents specifically, Alan Chan and colleagues argue for creating external agent infrastructure; that is, common protocols and guardrails to mediate how autonomous AI agents interact with one another and the world, including consistent agent identities, standardized communication rules, and mechanisms to detect or halt harmful agent actions at an infrastructural level (Alan Chan, Kevin Wei, Sihao Huang, Nitarshan Rajkumar, Elija Perrier, Gillian K. Hadfield, and Markus Anderljung, "Infrastructure for AI Agents," arXiv, arXiv:2501.10114, version 2, May 16, 2025).

The literature is clear that corporate governance has its limits. A central concern is whether profit-driven firms can sustain stringent safety practices that might conflict with short-term commercial interests. Traditional corporate law incentivizes maximizing shareholder value, which can discourage investments in ethics and safety that do not directly contribute to profit.¹⁷ Some frontier AI companies have experimented with nontraditional corporate structures to prioritize their mission—for instance, OpenAI was founded as a non-profit and later became a capped-profit hybrid, and Anthropic's governing documents center a public benefit mission. These structures aim to give boards more leeway to prioritize safety over profits. Yet OpenAI's past few years illustrate the fragility of such mechanisms: In late 2023, the company's board (composed of nonprofit trustees) abruptly fired the chief executive officer, Sam Altman, over transparency concerns, only to see an internal and investor backlash force a reversal within days.¹⁸ In 2025, OpenAI restructured by converting its commercial arm into a Delaware Public Benefit Corporation (OpenAI Group Public Benefit Corporation)—a for-profit structure—while its nonprofit parent (now the OpenAI Foundation) retained control; the recapitalization closed on October 28, 2025.¹⁹

The broader lesson is that voluntary corporate governance measures require supportive incentive structures. Market forces can undermine even sincere efforts at responsible AI development, unless there are countervailing rewards or requirements for safety. Some incentives are reputational (e.g., being seen as an “ethical AI leader” can attract business and talent) or anticipatory (preempting regulation by self-policing).

In summary, corporate self-regulation is a vital first layer of AI governance and has produced many promising practices, but it works best in tandem with external oversight and transparency. These limitations have driven growing interest in independent assurance mechanisms that can validate and reinforce internal efforts.

Independent Evaluation and Assurance

A growing body of work proposes instituting independent oversight and accountability mechanisms to complement companies' internal efforts. Some scholars have argued for third-party compliance reviews to verify that companies follow their declared AI safety frameworks.²⁰ Such independent reviews would function analogous to financial audits, providing assurance to the public (and regulators) that companies are adhering to best practices for high-stakes frontier AI models. Another proposal is instituting third-party red-teaming and flaw-disclosure processes: Shayne Longpre and colleagues suggest mechanisms in which external researchers can safely disclose vulnerabilities or dangerous behaviors in AI models to the developers and

¹⁷ Lynn A. Stout, *The Shareholder Value Myth: How Putting Shareholders First Harms Investors, Corporations, and the Public*, Berrett-Koehler Publishers, 2012.

¹⁸ Jeffrey Dastin and Aditya Soni, “Sam Altman to Return as OpenAI CEO After His Tumultuous Ouster,” Reuters, November 22, 2023.

¹⁹ Bret Taylor, “Built to Benefit Everyone,” OpenAI, October 28, 2025; Deepa Seetharaman, Stephen Nellis, and Deborah Mary Sophia, “Microsoft, OpenAI Reach Deal Removing Fundraising Constraints for ChatGPT Maker,” Reuters, October 28, 2025. Some commentators viewed this episode as evidence of *mission drift*; despite a novel governance model, market and market-like pressures (from investors, employees, and such partners as Microsoft) pushed the company back toward a profit-driven trajectory. In corporate governance terms, *amoral drift* can slowly erode a firm's ethical commitments as the firm chases funding and talent. A Harvard Law Review analysis pointed out that even with reduced shareholder rights, AI start-ups remain vulnerable to pressure from what it calls *superstakeholders*—e.g., venture investors, key employees with equity, or strategic partners—who have financial stakes and can effectively steer the company toward profit maximization. For more, see Harvard Law Review Association, “Amoral Drift in AI Corporate Governance,” *Harvard Law Review*, Vol. 138, No. 6, April 2025.

²⁰ Aidan Homewood, Sophie Williams, Noemi Dreksler, John Lidiard, Malcolm Murray, Lennart Heim, Marta Ziosi, Seán Ó hÉigeartaigh, Michael Chen, Kevin Wei, et al., “Third-Party Compliance Reviews for Frontier AI Safety Frameworks,” arXiv:2505.01643, version 1, May 3, 2025; Conrad Stosz, Karson Elmgren, Charles Foster, George Balston, Seth Donoughe, Samira Nedungadi, Michael Chen, Jasper Götting, Patricia Paskov, Sayash Kapoor, et al., *AEF-I: Minimum Operating Conditions for Independent Third Party AI Evaluations*, version 1, AI Evaluator Forum, undated.

oversight bodies rather than companies relying only on in-house testing.²¹ This is in line with the broader principle that “in-house evaluation is not enough” for complex general-purpose AI; there must be channels for outside scrutiny. Relatedly, Kevin Wei and Lennart Heim argue for well-designed AI incident-reporting systems that cover real-world harms and near misses and are informed by reporting regimes in safety-critical industries to enable ongoing, postdeployment external visibility and accountability for general-purpose AI systems.²²

Auditing, in particular, has been the subject of several recent proposals. The January 2026 report *Frontier AI Auditing* argues that transparency alone is insufficient and defines *frontier AI auditing* as rigorous third-party verification of safety and security claims that requires deep, secure access to nonpublic information. Such auditing also grades on AI Assurance Levels, from AAL-1 to AAL-4, to standardize how much confidence different audits can provide, from time-bounded system audits to continuous, deception-resilient verification.²³ Several authors have ties to new institutions that are aimed at building the infrastructure for AI auditing.²⁴ Notably, this work builds on earlier proposals, including Mökander and colleagues’ three-layered audit blueprint for AI systems, especially large language models (LLMs). Their framework calls for governance audits of AI developer organizations (to review corporate practices and controls), model audits of the AI system itself (testing the trained model for risks before deployment), and application audits of downstream uses of the model by external developers.²⁵ Auditing at all three levels in a structured and coordinated way is seen as a promising governance tool to identify and manage the ethical, safety, and legal risks of AI systems.²⁶

A growing set of organizations is building on the concept of third-party compliance assurance by turning standards into operational assurance. Emerging models pair technical criteria with insurance and other economic signals. In these models, certification against a defined standard can become a condition of insurability, and the information generated during certification can inform premium pricing and coverage terms. More broadly, insurers can act as private risk regulators; they have incentives to gather information about failure modes across clients, identify the controls associated with lower losses, and reward those controls through pricing, exclusions, deductibles, and underwriting requirements. For example, the Artificial Intelligence Underwriting Company (AIUC) links certification against its AIUC-1 agent standard with insurability for certain enterprise deployments,²⁷ illustrating how assurance and insurance could become mutually reinforcing. The Cloud Security Alliance has announced STAR (Security, Trust, Assurance, and Risk) for AI,²⁸ a registry-

²¹ Shayne Longpre, Sayash Kapoor, Kevin Klyman, Ashwin Ramaswami, Rishi Bommasani, Borhane Billi-Hamelin, Yangsibo Huang, Aviya Skowron, Zheng-Xin Yong, Suhas Kotha, et al., “Position: A Safe Harbor for AI Evaluation and Red Teaming,” *Proceedings of the 41st International Conference on Machine Learning*, July 2024.

²² Kevin Wei and Lennart Heim, “Designing Incident Reporting Systems for Harms from General-Purpose AI,” arXiv:2511.05914, version 1, November 8, 2025.

²³ Miles Brundage, Noemi Dreksler, Aidan Homewood, Sean McGregor, Patricia Paskov, Conrad Stosz, Girish Sastry, A. Feder Cooper, George Balston, Steven Adler, et al., *Frontier AI Auditing: Toward Rigorous Third-Party Assessment of Safety and Security Practices at Leading AI Companies*, January 2026.

²⁴ The AI Verification and Evaluation Research Institute is a nonprofit founded by Miles Brundage, the former head of policy research at OpenAI, and has the mission of making rigorous third-party auditing of frontier AI effective and universal. The institute is also a founding member of the AI Evaluator Forum, which has published minimum operating conditions for independent third-party AI evaluations.

²⁵ Jakob Mökander, Jonas Schuett, Hannah Rose Kirk, and Luciano Floridi, “Auditing Large Language Models: A Three-Layered Approach,” *AI and Ethics*, Vol. 4, No. 4, November 2024.

²⁶ Mökander et al., 2024.

²⁷ Artificial Intelligence Underwriting Company, homepage, undated.

²⁸ Cloud Security Alliance, “Cloud Security Alliance Launches STAR for AI, Establishing the Global Framework for Responsible and Auditable Artificial Intelligence,” press release, October 23, 2025.

based assurance program anchored to International Organization for Standardization (ISO) International Electrotechnical Commission (IEC) 42001 for AI management systems.²⁹ Independent certification vehicles and governance platforms—including the Responsible AI Institute and private audit providers—are likewise operationalizing life-cycle risk management and third-party attestation for enterprise buyers and deployers.³⁰

These independent evaluation and assurance mechanisms operate primarily at the level of individual organizations. Many AI risks, however, are systemic, cutting across firms, sectors, and borders, and require broader coordination to address.

Multistakeholder Coordination and Norm-Setting

A recurring theme in AI policy research is that managing AI risks requires coordination across many organizations and sectors, often through new networks or coalitions. Rather than a single global regulator for AI, some experts envision a distributed governance model—a network of networks—that involves governments, companies, academia, and civil society, each of which contributes to governance functions.³¹ The idea is that a decentralized, collaborative approach can be more adaptive and resilient in the fast-moving AI landscape, channeling expertise and resources where they are needed and providing redundancy if one node fails.³² Cameron Kerry and colleagues, for example, describe the emerging global AI governance architecture as a networked and distributed system and view the diverse, decentralized efforts underway as beneficial.³³ They note that because AI's challenges are immense and novel, no one government or institution has the capacity to handle them alone, but by networking multiple centers of effort across industry groups, standards bodies, nongovernmental organizations (NGOs), and international organizations, the community can mobilize more expertise and “mobilize diverse resources and initiatives” to respond to AI risks.³⁴

We see this principle in action through various multistakeholder initiatives, beginning a decade ago and continuing at the time of this writing. The Partnership on AI (PAI), launched in 2016, brings together leading tech companies and civil society organizations to jointly develop AI best practices on such issues as fairness, safety, and transparency.³⁵ Other examples include the Global Partnership on Artificial Intelligence,³⁶ the World Economic Forum's AI Global Alliance,³⁷ and the Frontier Model Forum (FMF),³⁸ an industry-led coalition spearheaded by several leading AI labs to collaborate on safety research and evaluation for cutting-

²⁹ Jim Reavis, “Announcing the AI Controls Matrix and ISO/IEC 42001 Mapping—and the Roadmap to STAR for AI 42001,” Cloud Security Alliance, August 20, 2025; International Organization for Standardization and International Electrotechnical Commission, “Information Technology—Artificial Intelligence—Management System,” ISO/IEC 42001:2023, 2023.

³⁰ Responsible AI Institute, homepage, undated.

³¹ Allan Dafoe, *AI Governance: A Research Agenda*, Centre for the Governance of AI, Future of Humanity Institute, University of Oxford, August 27, 2018.

³² Scott J. Shackelford and Rachel Dockery, “Governing AI,” *Cornell Journal of Law & Public Policy*, Vol. 30, No. 2, Winter 2020.

³³ Cameron F. Kerry, Joshua P. Meltzer, Andrea Renda, and Andrew Wyckoff, “A Network Architecture for Global AI Policy,” Brookings Institution, February 10, 2025. The authors describe a “networked and distributed” governance architecture.

³⁴ Kerry et al., 2025.

³⁵ Partnership on Artificial Intelligence, homepage, undated.

³⁶ Global Partnership on Artificial Intelligence, homepage, Organisation for Economic Co-operation and Development, undated.

³⁷ World Economic Forum, “AI Global Alliance,” webpage, undated.

³⁸ Frontier Model Forum, “Progress Update: Advancing Frontier AI Safety in 2024 and Beyond,” August 29, 2024.

edge AI models. Governments have also begun coordinating through the emerging AISI Network,³⁹ formally known as the International Network of AI Safety Institutes, which was launched at a convening in San Francisco in November 2024 to align technical work on AI safety research, testing, and evaluation, including engagement with AI developers, academics, and civil society leaders.

Multistakeholder collaboration also emerges on a more ad hoc basis, particularly in response to immediate AI risk concerns. Prominent examples include the Future of Life Institute's 2023 open letter, which called for a six-month pause on training the most powerful AI systems, and its 2025 letter, which called for the prohibition on the creation of superintelligent AI.⁴⁰ Although the proposed pause did not occur and there continues to be debate about the efficacy of open letters, these campaigns succeeded in injecting safety concerns into mainstream debate and spurred companies and policymakers to discuss new safeguards.

Comparative research suggests that industry-led safety initiatives can be effective if they are modeled on those in other high-stakes sectors. Morgan Simpson, Alejandro Ortega, and Robert Trager examine how voluntary safety programs in aviation and nuclear power improved outcomes by leveraging industry consortia.⁴¹ They conclude that similar efforts in AI can raise safety baselines ahead of regulation, provided that these efforts incorporate lessons on maintaining rigor and accountability without unduly stifling innovation.⁴²

Internationally, standardization organizations (e.g., ISO, the Institute of Electrical and Electronics Engineers [IEEE]) have also provided venues for businesses, academics, and civil society representatives to forge technical standards and ethics guidelines for AI. IEEE's Ethically Aligned Design standard and its series of AI standards (e.g., on transparency, bias mitigation) are notable examples of nongovernmental norm-setting driven by multistakeholder input. These standards often inform policy and corporate practice in tandem.⁴³ Because compliance is voluntary, norms can also diffuse and be tested before governments codify them; the European Union (EU) AI Act's approach of treating ISO/IEC harmonized standards as a compliance pathway is an example of this sequencing. That said, civil society's formal role in these processes has remained thin in practice. Research consistently finds that AI standards committees are dominated by large technology companies, and civil society, subject-matter experts (SMEs), and academic institutions are deterred by resource demands, opaque procedures, and the cost of accessing documents still under development.⁴⁴

³⁹ National Institute of Standards and Technology, "Fact Sheet: U.S. Department of Commerce & U.S. Department of State Launch the International Network of AI Safety Institutes at Inaugural Convening in San Francisco," November 20, 2024.

⁴⁰ Future of Life Institute, "Pause Giant AI Experiments: An Open Letter," March 29, 2023; Billy Perrigo and Tharin Pillay, "'Time Is Running Out': New Open Letter Calls for Ban on Superintelligent AI Development," *Time*, October 21, 2025. In March 2023, the institute's open letter calling for a six-month pause on training the most powerful AI systems was signed by thousands of tech leaders, researchers, and public figures. In 2025, the institute released another letter calling for the prohibition on the creation of superintelligent AI. A diverse array of stakeholders, including actors Joseph Gordon-Levitt and Stephen Fry, rapper will.i.am, and author Yuval Noah Harari signed their names to the letter.

⁴¹ Morgan Simpson, Alejandro Ortega, and Robert Trager, "Voluntary Industry Initiatives in Frontier AI Governance: Lessons from Aviation and Nuclear Power," AI Governance Initiative, Oxford Martin School, University of Oxford, January 20, 2025.

⁴² On coordinated restraint among frontier developers specifically, Jide Alaga and Jonas Schuett propose an evaluation-based coordination scheme in which developers would collectively agree to pause or take other joint actions if evaluations reveal dangerous capabilities, arguing that shared risk information can prevent races to the bottom on safety. See Jide Alaga and Jonas Schuett, "Coordinated Pausing: An Evaluation-Based Coordination Scheme for Frontier AI Developers," arXiv:2310.00374, September 30, 2023.

⁴³ Standards have fared reasonably well in fast-moving technology domains because they tend to be performance-based rather than prescriptive—setting governance outcomes rather than fixing methods—and are updated on a regular review cycle, which avoids locking in approaches that could quickly become obsolete. See IEEE, *Ethically Aligned Design: A Vision for Prioritizing Human Well-Being with Autonomous and Intelligent Systems*, April 2019.

⁴⁴ Christine Galvagna, *Inclusive AI Governance: Civil Society Participation in Standards Development*, Ada Lovelace Institute, March 30, 2023; Robert Kilian, Linda Jäck, and Dominik Ebel, "European AI Standards—Technical Standardisation and

The literature also suggests the potential efficacy of a co-regulatory approach; that is, leveraging these networks to craft norms and prestandards, which governments can later reinforce with law.⁴⁵ The EU's approach exemplifies this: While drafting the AI Act, the European Commission worked with industry and others to publish a voluntary AI Code of Practice on July 10, 2025, to guide general-purpose AI models.⁴⁶ Companies who sign onto this code commit to measures consistent with the act's aims (such as testing for discriminatory impacts), creating an interim compliance regime until the law is enforceable. This code was "co-designed by AI stakeholders" and is meant to prepare industry for forthcoming legal requirements.⁴⁷ Such interplay shows how multistakeholder outputs can directly inform and expedite formal governance.

The proliferation of these initiatives reflects broad recognition that AI governance cannot rest with any single actor. Yet coordination alone does not guarantee accountability, particularly when, as the standards literature notes, affected communities and public-interest organizations remain underrepresented in the processes that set the rules.

Civil Society Advocacy and External Pressure

Beyond formal frameworks and coalitions, civil society plays a critical watchdog and advocacy role in shaping AI's trajectory. NGOs, academics, media, and even tech workers have used transparency and public pressure to hold companies accountable and push for higher standards. Advocacy campaigns have directly led to changes in industry behavior. A notable case occurred in 2020, when major technology firms came under fire for selling facial recognition software to law enforcement despite the known bias and privacy risks. In the face of intense civil society and shareholder pressure—including studies exposing racial bias in these systems and petitions from civil rights groups—Amazon and Microsoft were effectively forced to halt sales of their facial recognition technology to police departments.⁴⁸ IBM went so far as to exit the facial recognition market entirely, citing ethical concerns. This example demonstrates how public outcry and NGO advocacy can press businesses into self-imposed moratoriums to prevent social harms.

Civil society organizations also contribute expertise and independent research that illuminates AI risks, often prompting industry responses. Transparency initiatives are another avenue: Such projects as the AI Incident Database (an open repository of AI failures and near misses that is maintained by researchers and volunteers) encourage organizations to report and learn from one another's mistakes, akin to safety reporting in aviation.⁴⁹ By documenting incidents, such as self-driving car crashes and algorithmic discrimination cases, these civil society-led efforts create public records that both regulators and companies can use to identify systemic problems. In the global context, coalitions of NGOs have influenced international AI policy debates—from calling for bans on autonomous weapons and predictive policing AI to shaping the human

Implementation Challenges Under the EU AI Act," *European Journal of Risk Regulation*, Vol. 16, No. 3, September 2025.

⁴⁵ Edward J. Balleisen and Marc A. Eisner, "The Promise and Pitfalls of Co-Regulation: How Governments Can Draw on Private Governance for Public Purpose," in David Moss and John Cisternino, eds., *New Perspectives on Regulation*, The Tobin Project, 2009; Michèle Finck, "Digital Co-Regulation: Designing a Supranational Legal Framework for the Platform Economy," *European Law Review*, Vol. 43, No. 1, 2018.

⁴⁶ European Commission, "General-Purpose AI Code of Practice Now Available," news release, July 10, 2025a.

⁴⁷ European Commission, *General-Purpose AI Code of Practice*, July 10, 2025b, chapter on safety and security, Measure 3.2, and Appendixes 1.1–1.3; European Commission, "Questions and Answers on the Code of Practice for General-Purpose AI," July 11, 2025c; European Commission, "Drawing-Up a General-Purpose AI Code of Practice," last updated March 23, 2026.

⁴⁸ Amazon News, "We Are Implementing a One-Year Moratorium on Police Use of Rekognition," news release, June 10, 2020; Jay Greene, "Microsoft Won't Sell Police Its Facial-Recognition Technology," *Washington Post*, June 11, 2020.

⁴⁹ AI Incident Database, "About," Responsible AI Collaborative, undated-a.

rights provisions in the EU AI Act.⁵⁰ Many governments now actively consult civil society organizations as stakeholders when developing AI strategies, recognizing the sector's valuable input on societal risks.⁵¹

The literature notes that for multistakeholder governance to truly benefit society, actors and institutions across civil society must have real influence and its representatives should not be merely token participants. Eva Erman and Markus Furendal explore how nonstate actors (such as expert NGOs) can gain legitimacy as *agents of democracy* even without formal public mandates.⁵² They argue that if these actors demonstrate moral authority (standing up for widely held values), epistemic authority (expert knowledge), or representative authority (speaking for affected communities), they contribute to democratizing AI governance in practice. Indeed, civil society can bolster what governance scholars call *input legitimacy* (inclusive participation in decisionmaking) and *output legitimacy* (ensuring that outcomes reflect societal values). Seen in this light, civil society is a cogovernor of AI, providing checks and balances that neither government nor industry alone reliably supply.

However, it is important not to romanticize civil society or treat nonstate actors as straightforward proxies for the public. Civil society organizations are internally heterogeneous and vary widely in expertise, representativeness, and accountability. Some groups primarily reflect the interests of particular constituencies or funders rather than broad public values; others lack the technical capacity to evaluate complex AI systems and might inadvertently amplify oversimplified or misleading narratives about risk. Civil society actors can be subject to capture and performative dynamics. Well-resourced organizations based in the Global North often dominate multistakeholder processes, marginalizing perspectives from less powerful communities and those from the Global South.⁵³ Companies might use carefully curated multistakeholder forums to legitimize predetermined positions—a form of ethics-washing in which civil society participation conveys democratic or moral authority without commensurate influence over outcomes.⁵⁴ Research on greenwashing suggests a similar pattern: Such behaviors proliferate where information asymmetries between firms and external observers are greatest, and verification mechanisms themselves can be captured as firms adapt by targeting intermedi-

⁵⁰ Campaign to Stop Killer Robots, “Urgent Action Needed to Ban Fully Autonomous Weapons: Non-Governmental Organizations Convene to Launch Campaign to Stop Killer Robots,” press release, April 23, 2013. See also Mary Wareham, *Stopping Killer Robots: Country Positions on Banning Fully Autonomous Weapons and Retaining Human Control*, Human Rights Watch, August 10, 2020.

⁵¹ Emerging work argues that nongovernmental actors should help ensure that AI's benefits are broadly shared and not just act to mitigate risks. Sumaya Nur Adan and colleagues propose mechanisms for redistributing AI-driven gains and transferring technology to underserved regions, while cautioning that current trajectories could exclude billions from AI's benefits. See Sumaya Nur Adan, Joanna Wiaterek, Varun Sen Bahl, Ima Bello, Camila Beltran, Tobias Dierks, Luise Eder, Liam Epstein, Abra Ganz, Natalie Kiilu, et al., *AI Benefit-Sharing Framework: Balancing Access and Safety*, AI Governance Initiative, Oxford Martin School, December 1, 2025. Claire Dennis and colleagues complement this by disaggregating benefit-sharing into concrete routes and emphasizing that civil society engagement is essential after commitments are announced to guard against coercive dependencies. See Claire Dennis, Sam Manning, Stephen Clare, Boxi Wu, Jake Okechukwu Effoduh, Chinasa T. Okolo, Lennart Heim, and Katya Klinova, *Options and Motivations for International AI Benefit Sharing*, Centre for the Governance of AI, Future of Humanity Institute, University of Oxford, February 6, 2025.

⁵² Eva Erman and Markus Furendal, “The Democratic Role of Non-State Actors in the Global Governance of Artificial Intelligence,” *Critical Review of International Social and Political Philosophy*, online ahead of print, 2025.

⁵³ Marie-Therese Png, “At the Tensions of South and North: Critical Roles of Global South Stakeholders in AI Governance,” *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency*, Association for Computing Machinery, 2022; Aubra Anthony, Lakshme Sharma, and Elina Noor, *Advancing a More Global Agenda for Trustworthy Artificial Intelligence*, Carnegie Endowment for International Peace, April 30, 2024.

⁵⁴ Mario D. Schultz, Ludovico Giacomo Conti, and Peter Seele, “Digital Ethicswashing: A Systematic Review and a Process-Perception-Outcome Framework,” *AI and Ethics*, Vol. 5, No. 2, April 2025.

aries.⁵⁵ These dynamics do not negate the value of civil society participation, but they underscore that civil society is a contested governance arena rather than an unambiguous corrective to state or corporate power.⁵⁶

Summary Findings from Semistructured Interviews

The semistructured interviews present an emerging picture of how practitioners in business and civil society perceive transformative AI risks, where those risks materialize, and which nonstate interventions seem feasible to interviewees in the near term. The interview protocol covered such topics as perceived AI risks in the near versus long term, existing practices for managing AI risk within organizations, views on what industry or civil society could do beyond the status quo, business and civil society's dependencies on government action, and concerns about feasibility or unintended consequences of nongovernmental interventions.

Interviewees consistently described a dual horizon of concern. On the near horizon, they emphasized ongoing harms—fraud, scams, and nonconsensual imagery—that already strain response capacity and are unevenly distributed. One interviewee warned, “What worries me about these [AI] systems is what is already happening—AI-generated child sexual abuse material. We are not even close to being able to field all of it,” noting that those “disproportionately affected are not in a position to participate.” Another observed that “scams are highly prevalent in Indonesia and India,” underscoring geographic disparities in realized harms. On the longer horizon, interviewees pointed to potential risks related to loss of control, concentration of power, and expanding “cognitive delegation.” Several anticipated “societal upheaval” and “labor displacement” and emphasized that “cognitive labor can be displaced,” and another noted that certain roles “might be able to be replaced . . . in 2 years” as the autonomy of AI systems increases.

Confidence in government preparedness was generally limited across interviewees. Interviewees attributed this to the pace of change and unsettled mandates that dampen internal incentives for sustained safety investment in corporations—a twin lack of expert consensus on transformative AI risks and clear regulatory mandates that makes it “difficult to create internal company incentives.” A few participants emphasized comparatively tractable subnational levers: ongoing work on an “AI whistleblower protection bill,” “liability reform . . . at the state level [e.g., Senate Bill 1047],” and coordinated strategies in which “state-level action—in conjunction with California and New York” can approximate a de facto national framework. Several also pointed to local infrastructure tools—“you can take advantage of zoning and permitting . . . data centers” to enforce certain rules, while cautioning that “companies can threaten not to do work in an area” and that “the state may preempt local” action.

⁵⁵ The greenwashing literature further warns, however, that verification mechanisms themselves can be captured: As independent certifications and standards become established, firms adapt by targeting those intermediaries, producing a cat-and-mouse dynamic between advocacy and corporate co-optation. For more on greenwashing, see Magali A. Delmas and Vanessa C. Burbano, “The Drivers of Greenwashing,” *California Management Review*, Vol. 54, No. 1, October 2011; A. W. Montgomery, Thomas P. Lyon, and Jasper Barg, “No End in Sight? A Greenwash Review and Research Agenda,” *Organization & Environment*, Vol. 37, No. 2, June 2024; Thomas P. Lyon, and A. Wren Montgomery, “The Means and End of Greenwash,” *Organization & Environment*, Vol. 28, No. 2, June 2015. Sports-washing scholarship offers an analogous lesson at the geopolitical level: Reputational laundering through positive brand association persists until civil society and media pressure raise the costs of symbolic participation sufficiently to demand substantive change. For more, see Michael Skey, “Sportswashing: Media Headline or Analytic Concept?” *International Review for the Sociology of Sport*, Vol. 58, No. 5, August 2023.

⁵⁶ Simone Chambers and Jeffrey Kopstein, “Bad Civil Society,” *Political Theory*, Vol. 29, No. 6, December 2001; John S. Dryzek, “Global Civil Society: The Progress of Post-Westphalian Politics,” *Annual Review of Political Science*, Vol. 15, 2012. Designing AI governance arrangements that rely on civil society therefore requires attention to such organizations’ internal accountability, their diversity of representation, their transparency regarding funding and mandates, and ensuring that there are mechanisms so that civil society actors are subject to appropriate checks and contestation. Ranjit Lall, “Making Global Governance Accountable: Civil Society, States, and the Politics of Reform,” *American Journal of Political Science*, Vol. 69, No. 1, January 2025.

Organizational readiness was described as uneven, and good intentions, interviewees suggested, were outpacing capacity, especially among smaller deployers. As one interviewee put it, “Most smaller [organizations] don’t have money for extra things . . . when there’s no regulatory teeth, no incentives, and no roadmap,” such organizations gravitate toward “mitigations that companies can control.” One interviewee mapped risk across multiple loci that call for distinct tools: the lab and R&D pipeline, high-risk deployments, government decisionmaking that is reliant on technical assessments, large-scale diffusion, end-user institutions, and the broader epistemic environment. “For risks that primarily sit in the lab [and] R&D pipeline (such as early-stage loss of control) . . . you need some lever over the labs,” whereas deployment-level harms are better addressed through operational safeguards and do not require lab engagement.

On incentives, several interviewees pointed to demand-side mechanisms—procurement standards, insurance and tort exposure, and ratings or other assurance pathways—as practical ways to raise baseline practices where regulation lags. At the same time, these interviewees described coordination frictions: attempts to “generate consumer signals . . . only work if consumers have options,” and prior efforts to “solve the collective action problem . . . have not worked.” As a result, there is emphasis on intermediaries that aggregate demand (e.g., sectoral consortia, large deployers) and pair purchasing signals with credible third-party assurance—essentially to “do the labs’ work for them so they can take it on.”

Many interviewees across sectors also expressed concern about labor displacement and downstream social effects. They anticipated concentrated shocks in particular sectors and regions and additional risks of wage pressure, job loss, and service-delivery strain. Several argued that workforce transition planning and community-level support should be treated as core elements of AI risk management rather than discretionary corporate social responsibility. One interviewee noted that the objective is to ensure “human thriving,” adding that organizations “need a level of human thriving to have a functional business model.”

A number of interviews referred to using operations to improve system safety. Interviewees stressed building good friction—permissioning, human-in-the-loop checks, and clear escalation routes that slow and instrument high-risk actions. Illustrative phrases included the following: “Throw up some wrenches . . . add some good friction . . . build [it] into operations,” “ask permission,” and “create friction between people and those who have power over technology.” Several also emphasized the importance of documented escalation: “Write down who can say yes, and when someone must say no,” and the need to “slow down the riskiest pathways without stopping routine work.”

Civil society capacity was characterized as energetic but fragmented and resource-constrained. One interviewee stated, “Funding is going down. Power is going down,” and recommended visible procedural steps to build legitimacy—“go for easy asks, build coalitions, outreach to media.” One interviewee underscored the role of public narratives and investigative journalism, akin to “the kinds of things we had around GMOs [genetically modified organisms],” in which “people like us don’t consume things like this” functioned as a norm-shaping refrain along with formal governance.

These findings reflect perceptions and experiences reported by stakeholders rather than consensus estimates or causal claims. They are intended to inform subsequent analysis of feasible nonstate interventions and the conditions under which those interventions are most likely to be effective.

Summary of Observations from RAND Games

Across five runs of the Business and Civil Society exercises (June–December 2025; more than 70 participants in the United States and United Kingdom), participants played themselves as leaders from business, civil society, philanthropy, academia, and adjacent sectors and responded to a two-turn crisis: an initial economic shock following the release of a frontier model, followed by a later turn featuring cascading infrastructure disruption, degraded communications, and uncertain attribution. The games were designed to elicit the

judgments, information needs, coordination challenges, and “no-regret” actions that become salient when events outpace normal governance rhythms.

Several themes recurred consistently across sessions:

- Participants treated frontier AI as simultaneously a national security risk and a social legitimacy crisis: Managing technical risk was viewed as inseparable from sustaining public cohesion, order, and purpose.
- Participants repeatedly emphasized the risk of an economy without participation, arguing that economic stabilization and social resilience must be addressed together through visible, credible mechanisms that sustain livelihoods and meaning (including sense of purpose) during rapid labor displacement.
- Leadership and narrative coherence were treated as decisive capabilities, particularly under conditions that included misinformation, deepfakes, and uncertainty. When communications degraded, participants prioritized access to trusted voices and a coherent, repeated public account of what was happening and the actions that were being taken.
- Participants stressed that communication redundancy is a form of civil defense. They called for analog and offline modes of official communication and verification, paired with civil society institutions that can act as localized truth relays.
- A recurring diagnosis was that speed itself has become a governance variable: Decision cycles measured in weeks or months were seen as incompatible with cascading failures unfolding in hours. This drove demands for preauthorized triggers, predelegated authorities, and prebuilt coordination networks that can activate rapidly without improvisation.
- Fiscal policy (including new tax models and transition funds) was viewed as central to stability, but participants also noted that there were trade-offs between equity and innovation, which reinforced the value of designing “default” approaches before labor displacement accelerates.
- Participants converged on the need for cross-sector and international coordination mechanisms, including crisis hotlines and structured compacts, while also flagging a moral hazard: When essential technical capabilities are concentrated in private hands, alignment requires reciprocal accountability rather than ad hoc bargaining in crisis.

Some key issues shaped decisionmaking in the sessions. In backcasting discussions, participants returned to seven categories of issues that repeatedly determined what information they sought and what actions they prioritized: (1) economic transition, employment, and legitimacy; (2) communication collapse and narrative control; (3) governance speed and the shrinking decision window; (4) fiscal tools and balancing equity with innovation; (5) cross-sector and international coordination; (6) civil resilience and basic needs; and (7) agency, helplessness, and psychological readiness.

The game-derived insights reinforce the need for nongovernmental action and were used, alongside the literature review and interviews, to help define a framework for business and civil society actors to envision roles and responsibilities they can play to help mitigate transformative AI risks.

Analytical Framework: Emerging Roles for Business and Civil Society in Managing and Mitigating Risk

A clear finding from the literature, our interviews, and the games is that business and civil society *can* play critical roles in transformative AI risk management and governance. By *roles*, we mean clusters of responsibilities and activities that these actors are well suited to perform because of their positions in the AI develop-

ment and deployment life cycles, as well as in society. These roles emerged from our research as distinct but interlocking areas where action is both useful and feasible under existing conditions.

We identify three roles for business and civil society actors. The first is managing technical and operational risks: the work of ensuring that AI systems are built, tested, and deployed with adequate safeguards. The second is shaping market and business incentives: using procurement, investment, insurance, and industry norms to make safety a winning strategy. The third is supporting social stability and public trust: cushioning the workforce disruptions, community shocks, and legitimacy strains that transformative AI will generate. The roles are distinct but reinforcing; for example, technical controls reduce the hazards that market mechanisms must price. In what follows, we describe in more detail the kinds of actions and responsibilities that actors across the AI ecosystem can take to mitigate safety and societal risks of transformative AI (Table 1).

Before proceeding, three caveats merit emphasis. First, many of the measures that we highlight are not wholly novel. Frontier labs, deployers, and AI safety institutes already perform elements of rigorous pre-deployment testing, incident-sharing, and regulator-lab collaboration through bilateral memorandums. Our contribution is to generalize those practices beyond a small set of frontier actors, specify minimum baselines

TABLE 1
Actors and Possible Roles Across Business and Civil Society

Actor Type	Positioned to Lead On	Minimum Baseline Actions Now
Frontier developers and model providers	• Model design • Evaluation • Release • Access controls	• Dangerous-capability evaluations • Documented safety cases • Staged release • Usage controls • Incident logging • Break-glass procedures
Deployers and integrators	• Use-case governance • Monitoring • Human oversight	• Inventory AI uses • Risk-triage use cases • Predeployment sign-off for consequential uses • Override and escalation procedures
Upstream infrastructure and dependency providers	• Supply-chain resilience • Systemic risk reduction	• Redundancy • Vendor security assessments • Access controls
Market intermediaries (e.g., buyers, insurers, investors, auditors, associations)	• Diffusing safety expectations through markets	• Procurement clauses • Due-diligence questionnaires • Assurance requirements • Eligibility criteria for coverage or investment
Technical and research institutions	• Independent evidence generation and evaluation	• External testing • Benchmarking • Red-teaming • Incident analysis
Standards, professional, and accountability organizations	• Translating norms into repeatable practice	• Incident taxonomies • Checklists • Templates
Worker, community, and service organizations	• Detecting downstream harms and supporting adaptation	• Workforce transition planning • Community resilience partnerships
Public-interest, journalistic, and advocacy organizations	• Surfacing harms • Representing affected groups • Shaping expectations	• Watchdog reporting • Incident documentation • Public pressure for higher standards

that are actionable for nonfrontier firms and civil society, and articulate how these practices can interface with government action.

Second, not all recommended actions should be pursued at once. A practical way to sequence them is to distinguish among three layers of effort: minimum baseline actions that organizations can begin now with existing authorities and modest resources; coordinated sectoral actions that require shared standards, pooled infrastructure, or third-party verification; and enabling public policy actions that are likely to matter over time but depend on government participation. This sequencing matters because feasibility varies sharply across actor types. Smaller deployers will often need lightweight checklists, shared tools, and sectoral support before they can adopt more-demanding practices that might be appropriate for frontier developers, cloud providers, or large regulated enterprises.

Third, none of these nongovernmental roles is a substitute for government action; rather, these roles bring down risk in the near term and inform smarter regulation in the longer term. Many interviewees and game participants expressed skepticism that governments will move quickly enough, which is why these nongovernmental efforts are so critical. Over the longer term, the aim is to have a system in which voluntary and market-driven measures work alongside formal oversight. Our analytic framework therefore also considers the interfaces to government for each role; that is, points at which private and civic initiatives can support and be reinforced by public policy.

Role 1. Managing Technical and Operational Risks

In this role, businesses take primary responsibility for preventing and controlling the technical risks of AI systems, serving as the first line of defense against accidents, misuse, or system failures. Civil society plays a complementary oversight role, offering external accountability, independent expertise, and pressure-testing of industry measures. The aim is to embed a culture of safety and accountability within companies developing and deploying AI and to ensure that critical safeguards exist before external enforcement steps in.

Companies developing and deploying AI are closest to where risks originate—in model training, system design, and real-world implementation. They have the knowledge and access needed to identify and mitigate issues, such as flawed training data, model vulnerabilities, or unsafe deployment contexts, before those issues lead to harm.

Organizations need a first-line approach to AI risk because many of the most preventable harms arise where models are designed, trained, integrated, and deployed. Interviews and game observations highlight a spectrum of risks: ongoing, unevenly distributed harms in the near term, such as scams, nonconsensual imagery, and operational misuse, and concerns related to loss of control, concentration of power, and rapid cognitive delegation over the longer term. Practitioners also reported that intent is often outpacing capacity, especially at smaller deployers. What they described as *good friction*—permissioning, human-in-the-loop review, and clear escalation pathways—can slow the riskiest pathways without stalling routine work. At the same time, the governance expectations appropriate for a frontier model developer are not identical to those appropriate for a downstream adopter or resource-constrained organization. The level of formality should scale with capability, deployment context, and level of control over the system. Frontier developers, cloud and infrastructure providers, and high-risk deployers in such sectors as health, finance, and critical infrastructure should maintain distinct AI risk functions, formal escalation routes, and documented safety cases.

Smaller deployers and downstream adopters should instead begin with a thinner baseline: named executive ownership, an inventory of AI uses, use-case risk triage, human-override procedures for consequential applications, and access to external expertise through sectoral associations, auditors, insurers, or shared service providers. Observations from games also surfaced a crisis-readiness gap. One game mantra—“panic responsibly”—captures the need to move early where resilience compounds while avoiding brittle or performative responses. Taken together, these observations reinforce the case for auditable governance artifacts,

pre- and postdeployment testing, confidential hazard reporting, and break-glass drills as standard operating practice, but with implementation scaled to actor type and risk.

Key actions could include the following:

- **Build internal governance and evaluation infrastructure.** Organizations should right-size governance for the risks they create or control. For frontier developers and high-risk deployers, this could mean formal board or executive oversight of AI risk, a defined risk appetite,⁵⁷ independent second-line review, internal or external audit,⁵⁸ documented safety cases for high-impact systems, and clear disclosure and escalation protocols. For example, boards could mandate quarterly AI risk reports that include an inventory of existing AI systems, the extent of testing and validation each system has undergone, any incidents or near misses that have occurred, and mitigation status for identified issues. Organizations could also ensure that disclosure protocols are in place so that if a major AI incident occurs, the company can communicate responsibly to stakeholders and possibly regulators. Putting AI on the board's agenda will not fix all risks—any more than it has for cyber or financial risk—but building governance infrastructure does make it clearer who is responsible and helps ensure that AI safety is treated as a regular part of major business decisions. For smaller deployers, the minimum viable version is more feasible; that is, designate a responsible executive, maintain an inventory of existing AI systems and use cases, classify those uses by risk, require predeployment sign-off for consequential applications, and establish a documented override and incident-escalation process.

Across sectors, trade associations, professional bodies, and standards organizations can lower implementation costs by offering common templates for inventories, incident taxonomies, testing checklists, and model or application review forms. The goal is not bureaucratic uniformity but rather to ensure that organizations know what systems they are using, what risks those systems create, who is accountable for those systems and their performance, and what evidence is required before deployment or scale-up.

High-risk AI systems should undergo predeployment testing and postdeployment monitoring. Many frontier developers already conduct some combination of internal red-teaming, staged releases, and external security testing. The step that we propose is to turn these ad hoc practices into an auditable minimum standard: tying tests to explicit go or no-go criteria, documenting those tests as part of a safety case, and extending them beyond a handful of labs to any actor deploying models that could plausibly generate systemic risk. A structural caveat applies: Predeployment evaluations face recognized adequacy limits. General-purpose models can exhibit emergent behaviors following deployment that were not detectable during testing, and evaluation protocols can be gamed by sufficiently capable systems.⁵⁹ Given these limitations, a clean evaluation is a necessary but not sufficient assurance and should be paired with robust postdeployment monitoring and the operational capacity to rapidly update or retract systems.⁶⁰

⁵⁷ Cibralic et al., 2025.

⁵⁸ Schuett, 2025. For more of the Three Lines of Defense model, see Institute of Internal Auditors, *The Three Lines of Defense in Effective Risk Management and Control*, January 2013.

⁵⁹ Yoshua Bengio, Stephen Clare, Carina Prunkl, Maksym Andriushchenko, Ben Bucknell, Malcolm Murray, Shalaleh Rismani, Conor McGlynn, Nestor Maslej, Philip Fox, et al., *International AI Safety Report 2026*, February 2026; Anita Rao, Drew Keller, Neha Kalra, Ryan Steed, Kweku Kwegyir-Aggrey, Kevin Klyman, Diane Staheli, and Stevie Bergman, *Challenges to the Monitoring of Deployed AI Systems*, Center for AI Standards and Innovation, National Institute of Standards and Technology, NIST AI 800-4, updated March 18, 2026.

⁶⁰ The Center for AI Standards and Innovation at the National Institute of Standards and Technology is exploring ways to improve postdeployment monitoring of AI systems and is a strong partner for industry actors interested in learning more

- **Establish incident- and hazard-reporting mechanisms.** AI governance will remain reactive unless organizations can report and learn from incidents and near misses in a structured way. In the near term, sector-specific reporting channels—whether through industry consortia, professional bodies, insurers, or trusted third-party organizations—can help firms share lessons on failures, misuse patterns, and unsafe model behavior. But private reporting alone is unlikely to be sufficient. Insurers may collect incident information across policyholders, but there are structural reasons to be skeptical that they will voluntarily pool or publish those data.⁶¹ NGO-led databases are valuable for transparency but often under-capture near misses and lack the operational detail needed for technical learning.

Over time, governments likely will need to facilitate or mandate standardized disclosure for at least some categories of serious incidents. The history of cyber insurance is instructive here: despite years of voluntary data-sharing initiatives among insurers and information-sharing and analysis centers, it took government intervention through breach notification mandates and regulatory guidance to generate the data infrastructure that the market needed to have the information to price risk credibly. AI might follow a similar trajectory. A promising intermediate model would be a voluntary, anonymized reporting mechanism supported by a public-interest or government-backed institution—analogue to aviation near-miss systems—that protects reporters while enabling cross-organizational learning.

A model worth examining specifically is the National Aeronautics and Space Administration's Aviation Safety Reporting System,⁶² which has demonstrated for decades that a government-administered, confidentiality-protected, nonpunitive reporting system can generate the near-miss data that markets, on their own, do not produce. The National Institute of Standards and Technology would be a natural institutional home for an analogous AI safety reporting system, consistent with recommendations from the Federation of American Scientists.⁶³ Industry-led solutions have worked in some cases—the Institute of Nuclear Power Operations achieved remarkable safety improvements in commercial nuclear power through a member-owned confidential reporting structure⁶⁴—but these required strong external pressure to establish, and they represent exceptions rather than the norm.

- **Invest in safety research and tools.** Many AI risks (especially frontier risks) are systemic, cutting across organizations; no single company can fully address them in isolation. Our interviews and games repeatedly raised the theme that certain safety solutions should be developed collaboratively as public goods. Examples include methods for adversarial testing of AI models, libraries of known vulnerabilities and corresponding mitigations, and standard benchmarks for evaluating model behavior on safety-relevant dimensions.

(National Institute of Standards and Technology, "Center for AI Standards and Innovation (CAISI)," webpage, undated).

⁶¹ Incident intelligence is a competitive asset, and sharing it erodes the informational advantage that allows a skilled insurer to price risk more accurately than rivals. See Daniel Schwarcz and Josephine Wolff, "The Limits of Regulating AI Safety Through Liability and Insurance: Lessons from Cybersecurity," *Fordham Law Review*, 2026.

⁶² National Aeronautics and Space Administration, "Aviation Safety Reporting System (ASRS): Program Briefing," webpage, undated.

⁶³ John Croxton, David Robusto, Satya Thallam, and Doug Calidas, "Message Incoming: Establish an AI Incident Reporting System," Federation of American Scientists, September 26, 2024.

⁶⁴ John E. Gudgel, "Insurance and the Public-Private Management of Risk at US Commercial Nuclear Power Plants," *Risk Management and Insurance Review*, Vol. 26, No. 4, Winter 2023; Cristian Trout, "Liability and Insurance for Catastrophic Losses: The Nuclear Power Precedent and Lessons for AI," *Generative AI and Law Workshop at the International Conference on Machine Learning*, 2024.

A historical analogy is the creation of the Common Vulnerabilities and Exposures (CVE) database in cybersecurity, which provides a shared index of software vulnerabilities across the entire industry.⁶⁵ In AI, one could imagine a shared repository of “known dangerous capabilities” or scenario playbooks for AI failure modes. Similarly, large-scale red-team exercises or challenges could be run jointly by multiple labs or with external experts, pooling knowledge on how to break (and thereby fix) advanced models. Industry consortia, academic partnerships, and philanthropic funding can support these pre-competitive efforts. The benefit is twofold: It lowers the cost barrier for smaller firms to implement safety (because tools and best practices are openly available), and it creates uniform expectations so that cutting corners becomes evident. To prevent this collaborative approach from being mere optics, there should be external validation of its outputs—e.g., academic peer review of shared safety tools and agreed metrics to track adoption (so one can see if, say, 90 percent of leading AI firms have implemented the top ten safety practices).

- **Establish “break-glass” safety controls and rehearsals.** For AI systems deployed in high-risk contexts, organizations could predefine emergency measures—the equivalent of fire alarms and extinguishers. This could include identifying specific triggers or red flags that would warrant halting an AI system (for instance, abnormal behavior that crosses a threshold) and assigning clear decision rights for who can pull the plug. Possible actions in a break-glass plan might be activating a capability throttle (limiting the AI’s operation to a safe mode), deploying a kill switch to shut it down or sever network access, rolling back to a known safe version of a model, or limiting application programming interface (API) rates if misuse is detected.⁶⁶

These plans should exist not only on paper but be rehearsed, much like disaster drills. Teams could simulate a scenario in which the AI goes awry and practice the shutdown or containment procedure. Additionally, it could be useful to maintain tamper-evident logs and conduct postincident reviews whenever these controls are triggered to learn and improve for the future. A culture of safety drills will improve readiness for events that today might seem unlikely but could have significant consequences if they catch organizations unprepared.

- **Secure the AI supply chain and critical infrastructure.** Transformative AI relies on complex infrastructure, from cloud computing data centers and specialized AI chips to the integration of AI systems into such critical domains as health care, finance, energy, and transportation. A failure or compromise in these areas could have cascading effects.

Participants in our study pointed to such scenarios as an AI system widely used for electric grid control failing unpredictably, or a malicious actor using AI to disable cloud infrastructure. The analogy was made to the 2003 Northeast U.S. blackout, in which a minor technical issue cascaded into a massive power outage.⁶⁷ To guard against such systemic failures, businesses should harden their AI supply chains and infrastructure. This includes ensuring redundancy (no single points of failure in cloud services), rigorous vendor security assessments (because many AI systems incorporate third-party models or datasets), and access controls to prevent unauthorized manipulation of AI models (such as by insider threat or external hacking).

⁶⁵ CVE Program, “Overview: About the CVE Program,” webpage, undated; MITRE, “CVE Program Celebrates 25 Years of Impact in Cybersecurity,” news release, October 22, 2024.

⁶⁶ Whether advanced AI systems can reliably be interrupted or constrained is, itself, a technically contested question. Some literature on corrigibility suggests that sufficiently capable systems might develop instrumental incentives to resist modification or shutdown. See Nate Soares, Benja Fallenstein, Stuart Armstrong, and Eliezer Yudkowsky, “Corrigibility,” *AAAI Workshops: Workshops at the Twenty-Ninth AAAI Conference on Artificial Intelligence*, 2015.

⁶⁷ U.S.–Canada Power System Outage Task Force, *Final Report on the August 14, 2003 Blackout in the United States and Canada: Causes and Recommendations*, April 2004.

Leading AI firms and cloud providers could also collaborate on contingency planning—for instance, agreeing on fallback procedures if a widely used AI service needs to be swiftly taken offline for safety reasons. From the civil society side, independent watchdogs could monitor concentration risks (e.g., if too much critical AI capability resides in one or two companies) and campaign for transparency around infrastructure safety. In other high-risk industries, NGOs have played a role in monitoring facilities (e.g., groups that keep tabs on nuclear plant safety). A similar approach could be applied where third-party auditors conduct AI infrastructure audits to ensure that, for example, hospitals using AI diagnostics have manual override processes or that autonomous vehicle networks have fail-safes if cloud connectivity is lost. This external check reinforces industry efforts and provides public reassurance that critical systems will not all collapse because of an AI glitch or cyberattack.

Barriers vary by sector. Regulated sectors already possess compliance, vendor-management, and continuity-planning functions that can absorb some of this work. By contrast, smaller firms may lack both leverage over upstream vendors and the internal expertise to evaluate AI supply-chain risks. For those actors, a realistic first step could be to adopt sector-issued supplier questionnaires, minimum contractual requirements, and contingency plans for rollback or manual fallback rather than attempting bespoke supply-chain assurance from scratch.

Conditions for success: For Role 1 initiatives to make AI safer, broad adoption and sincerity are key. One risk is fragmentation—if only a few leading companies implement strong internal governance while others (especially smaller start-ups) do little, then significant portions of AI activity remain vulnerable. Collaboration and capacity-building are needed to spread best practices across the industry, possibly through open-source toolkits or industry training programs for AI safety management.

Another concern is dominance by large firms. If only the biggest players shape the standards (even voluntarily), it could sideline smaller innovators or lead to “fox guarding the henhouse” situations. Involvement of nonindustry actors, such as academia, standards bodies, or civil society, in designing frameworks can mitigate this concern. Safety-washing was explicitly cautioned against by interviewees—companies might create safety boards or publish AI principles as public relations exercises without changing actual behavior. To counter this, internal measures should be paired with transparency and external auditability. Civil society can play a watchdog role. For example, if a company claims to have a robust risk management process, NGOs or journalists might later inquire about a known incident and how the company’s process handled it. If it emerges that the incident was ignored, the gap becomes public. Finally, incentives still matter even for voluntary safety; executives need to see that good AI governance aligns with business success (or at least, bad governance aligns with business risk). The next role addresses how to create those incentives.

Role 2. Shaping Business and Market Incentives

This role involves using business, market, and network levers to encourage responsible AI behavior across the private sector, especially in the absence of immediate regulation. Governments hold ultimate regulatory power (the big stick), but businesses and civil society can craft interim carrots and sticks through contractual requirements, industry standards, financial signals, and partnerships. The premise is that by altering the cost-benefit equation—making it in companies’ economic and reputational interest to prioritize safety—one can accelerate the adoption of safeguards.

Interviewees consistently pointed to tools on the demand side—procurement requirements, insurer and investor scrutiny, and credible ratings or assurances—as practical ways to raise minimum safety standards. Game discussions complemented this with concrete no-regret moves: independent security evaluations of leading labs, shared playbooks and standardized definitions for loss-of-control events, and decentralization to reduce single points of failure. These insights suggest that safety incentives are more likely to work

when buyers pool their purchasing power, compliance can be independently checked, and smaller firms have affordable ways (for example, open tools or subsidized evaluations) to meet the same standards.

Key actions could include the following:

- **Integrate safety into procurement and investment decisions.** Procurement was identified as one of the strongest levers. Large organizations that purchase or deploy AI solutions—whether they are corporations buying AI software, government agencies contracting AI systems, or even dominant tech platforms sourcing AI models—can set requirements in contracts that vendors must meet certain safety criteria. For example, a financial services firm procuring an AI risk-management tool could require the vendor to provide a model risk assessment and an audit report from an independent evaluator. If major customers in finance, health care, or defense start doing this, AI suppliers could begin to treat those safety steps as mandatory for winning business.

There is historical precedent. The U.S. military and the National Aeronautics and Space Administration, in the early aerospace industry, used procurement standards to greatly improve aviation safety: Contractors had to meet reliability benchmarks to get contracts.⁶⁸ At the time of this writing in early 2026, governments are attempting to use similar strategies for cybersecurity (e.g., requiring cloud providers to adhere to Federal Risk and Authorization Management Program [FedRAMP] security controls to sell to government).

For AI, individual companies—as well as a consortia of buyers—could form buyers' clubs to pool their market power and demand safer AI products.⁶⁹ For instance, a consortium of hospital systems could agree not to buy any AI diagnostic tool that has not undergone a robust clinical safety evaluation. Civil society can assist by ensuring that these efforts remain inclusive and fair—for example, watchdog groups might monitor that a buyers' consortium does not become a closed cartel that benefits only its members or raises entry barriers.⁷⁰

On the investment side, venture capital and institutional investors are increasingly attentive to AI governance. Investment due diligence can include an AI safety checklist, and investors might favor companies that proactively implement risk management (seeing it as indicative of a well-run company that avoids scandals). Some interviewees suggested that insurance underwriters and institutional investors could even team up to create a rating system for AI governance—akin to credit ratings or environmental, social, and governance (ESG) ratings—that scores companies on their safety practices. This would convert otherwise invisible governance quality into a tangible factor in business transactions.

Although the standardization problems with ESG offer a cautionary tale—companies self-report on inconsistent metrics, raters apply different weighting schemes that produce different scores for the

⁶⁸ Department of Defense Standard Practice MIL-STD-882E, *System Safety*, incorporating change 1, U.S. Department of Defense, September 27, 2023; National Aeronautics and Space Administration, *Reliability Program Requirements for Aeronautical and Space System Contractors*, NHB 5300.4(1A-1), January 1987.

⁶⁹ World Economic Forum, *Unlocking Public Sector AI: AI Procurement in a Box: AI Government Procurement Guidelines*, toolkit, 2020; Office for Artificial Intelligence, *Guidelines for AI Procurement*, United Kingdom government guidance, June 8, 2020.

⁷⁰ One structural constraint deserves explicit acknowledgment: Coordinated purchasing refusals by competing buyers can constitute a group boycott under Section 1 of the Sherman Act, depending on market structure and exclusionary effect. Buyers' club arrangements should be designed with antitrust counsel from the outset, and particular attention should be paid to membership openness, transparent and objective criteria, and scope limitations that target safety standards rather than competitor exclusion. Moreover, standards used in procurement should be transparent and open so that smaller AI producers know how to comply and civil society can critique or improve the criteria (U.S. Code, Title 15, Section 1, Trusts, Etc., in Restraint of Trade Illegal; Penalty [Sherman Act]).

same firm, and governance quality is inherently difficult to measure through observable proxies⁷¹—they also point to design principles that, if adopted, could help make AI safety ratings better. Ratings should be anchored to specific, independently verifiable practices rather than self-reported governance claims; methodologies should be standardized enough that different evaluators applying the same criteria reach comparable conclusions;⁷² and the institutions that develop and maintain ratings should be structurally independent of the firms they evaluate. There should also be some accountability mechanism if they certify false or misleading statements.

- **Develop liability and insurance frameworks that reward safety.** Liability and insurance can help internalize AI risk, but their effects will depend on market structure and information availability. In the near term, the most plausible role for insurance is not to solve all frontier AI risks but to reward observable controls in enterprise deployments, including logging and traceability, human-override mechanisms, secure model access, incident response plans, and independent evaluation for high-consequence uses. Firms that can demonstrate these practices could obtain coverage on better terms; firms that cannot could face exclusions, higher deductibles, or higher premiums.

Over time, stronger liability rules and mandatory insurance in selected applications might become appropriate, but it would be premature to assume that insurance markets will spontaneously solve coordination or catastrophe-risk problems on their own. In some domains, government might need to catalyze the market by, for example, clarifying liability, standardizing incident reporting, or signaling which forms of third-party assurance will count as evidence of due care. For smaller firms, pooled insurance mechanisms or sectoral mutuals might prove more feasible than bespoke coverage. The core principle is to align the cost of AI risk more closely with those who create or control it so that investments in safety become economically rational rather than purely voluntary.

In interviews, participants noted that insurers might require evidence of verifiable safety measures. This creates a financial reward (lower insurance costs or even the ability to get insured at all) for doing the right things.⁷³ Civil society can contribute by advocating that insurance and liability solutions do not exclude smaller players or lead to unfair outcomes. For example, if only big companies can afford the best safety certifications to lower their insurance, smaller start-ups might be priced out of markets. One remedy is pooled insurance or subsidized evaluations for smaller firms, possibly with philanthropic support. Additionally, NGOs and consumer advocates can push for liability reforms (some interviewees mentioned relevant ideas, such as an “AI whistleblower protection” law or state-level liability tweaks) to make it easier for harmed parties to get redress, which, in turn, pushes companies toward prevention. The key principle is aligning the costs of AI risks with those who create the risks so that safety investments become a rational choice.

Several near-term constraints on liability-based mechanisms deserve acknowledgment. First, existing U.S. tort doctrine makes harm attribution in AI supply chains difficult. The combination of multi-party development pipelines, the challenge of establishing proximate causation for probabilistic model

⁷¹ Florian Berg, Julian F. Kölbel, and Roberto Rigobon, “Aggregate Confusion: The Divergence of ESG Ratings,” *Review of Finance*, Vol. 26, No. 6, November 2022.

⁷² Notably, Berg, Koelbel, and Rigobon (2022) find that ESG rating divergence stems less from data scarcity than from differences in conceptual scope—what counts as “governance quality” varies across raters in ways that produce near-zero cross-rater correlation for governance dimensions even when financial risk scores converge. AI safety ratings face an analogous challenge: “Safety” is underspecified enough that different evaluators may be measuring different constructs, and this problem should be addressed through shared operationalization before any rating system is launched. For more, see Berg, Kölbel, and Rigobon, 2022.

⁷³ Cristian Trout, “Healthy Insurance Markets Will Be Critical for AI Governance,” *Lawfare* blog, December 17, 2025; Risk & Insurance, “Cyber Insurance Provides Both Carrot and Stick for Cybersecurity,” July 11, 2024.

outputs, and the statutory protections that could limit platform liability mean that injured parties face significant structural barriers to recovery. Second, the domestic political environment has contracted the near-term likelihood of federal liability reform. The Trump administration rescinded the prior executive framework on AI safety standards and has signaled opposition to AI-specific federal mandates.⁷⁴ It has also moved to limit interstate coordination on AI governance, reducing the de facto federal standard that state coalitions might otherwise approximate. Third, AI insurance markets remain nascent. Underwriters lack actuarial data to price AI-specific risks confidently, particularly for frontier model behavior, which means that premium incentives are likely to be modest and concentrated in enterprise applications with well-understood risk profiles. These constraints do not negate the longer-term importance of liability and insurance as governance levers, but they set a more realistic near-term baseline. The most achievable early steps operate within existing legal authority: contractual warranties and indemnification clauses in enterprise procurement; sector-specific liability frameworks in already-regulated domains, such as health care and financial services, in which existing law already assigns duties of care; and targeted state-level tort reforms in jurisdictions where the political conditions permit.

- **Form multistakeholder coalitions for norms and standards (including internationally).** Many AI risks are global—an unsafe model deployed on the internet can reach users worldwide, and competitive pressures span borders. Waiting for all governments to agree on treaties may be slow. Instead, coalitions of companies, NGOs, and experts can forge interim norms. As we discussed in the literature review, multistakeholder coalitions can fill governance gaps where no single government can impose rules. The actionable step is to move from general norm-setting to specific commitments. Companies and civil society could jointly agree on evaluation protocols for any model above a certain capability threshold and commit to not release models that fail certain tests. Another concrete area is to facilitate cross-border information-sharing on AI incidents, which could be managed by a trusted network of organizations across the globe.

Over time, these private and civil norms often influence or become incorporated into public international frameworks. For instance, if a de facto industry standard for AI auditing emerges and is adopted by dozens of major firms globally, regulators might later codify that standard or at least recognize compliance with it as evidence of due care. Civil society networks also help bring voices from the Global South and other underrepresented regions into what might otherwise be a club of big tech firms. This inclusivity is crucial to avoid a governance gap where only a few countries' perspectives shape AI norms.

Conditions for success: To shape incentives effectively, collective action is needed—one buyer or one insurer alone has limited sway, but coordinated groups can shift market expectations. This means forming alliances (e.g., buyers' consortia, multicompany pledges) and possibly getting some public-sector endorsement (even if not a mandate) to give efforts weight. Another condition is verification. Incentives work only if claims of safety can be trusted. That is why third-party assessments and certifications become important: They provide the credible signal that, for example, a vendor meets the procurement criteria or a firm deserves the insurance discount.

Civil society and academia can serve as the independent evaluators or standard-setters to ensure the integrity of these incentive structures. Additionally, a conscious effort must be made to include smaller com-

⁷⁴ Executive Order 14179, "Removing Barriers to American Leadership in Artificial Intelligence," Executive Office of the President, January 23, 2025, which rescinded Executive Order 14110, "Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence," Executive Office of the President, October 30, 2023. The January 2025 order directed agencies to review and revise or rescind AI-related guidance that is inconsistent with its policy of removing "undue burdens" on AI development.

panies and start-ups. If only large corporations can afford compliance, we risk a two-tier system. These voluntary or market-driven measures should ideally create a baseline that makes eventual regulation easier and more evidence-based, because by the time laws are written, there is already practical knowledge of what works (or what does not).

Role 3. Supporting Social Stability and Public Trust

This role addresses the societal and human impacts of transformative AI. Recognizing that AI's risks are not only technical malfunctions but also economic, institutional, and civic disruptions, this role calls on businesses and civil society to cushion shocks and to also help shape how societies could adapt. That includes mitigating labor-market upheaval, sustaining community resilience, maintaining informed public discourse, and ensuring that the benefits of AI are broadly shared. But it should not be understood as merely backfilling government capacity or repairing damage after the fact. Its broader purpose is to help shape a more durable social contract around AI: What forms of work, participation, and local institution-building remain essential? What new forms of contribution and human agency should be cultivated? And how can legitimacy be maintained during periods of rapid technological change? If Role 1 is about safety engineering and Role 2 about economic and market levers, Role 3 is about ensuring that adaptation to AI remains pro-human, publicly intelligible, and socially legitimate.⁷⁵ (Legitimacy here operates in two complementary senses: input legitimacy and output legitimacy. Both matter. Governance that is perceived as procedurally inclusive but ineffective will erode trust just as governance that produces good outcomes through opaque processes will prove fragile.⁷⁶)

Participants in our research widely agreed that the societal consequences of AI, such as job losses, inequality, erosion of trust, could be as destabilizing as any technical failure. We have already seen early signs, including artists protesting AI-generated content that threatens their income, educators worried about AI in schools, and workers in some sectors facing replacement by automation. If such issues are left unaddressed, they can lead to political backlash, social unrest, or the outright rejection of potentially beneficial technologies. Businesses have a direct stake here. An AI company's deployment could be halted by public outcry or its reputation ruined by perceptions of callousness. Civil society groups, for their part, are often on the ground helping those who fall through the cracks. In previous technological revolutions, proactive measures, such as the creation of pension plans or retraining programs during industrialization, helped ease such transitions and avoid worst-case social conflict. The same logic applies to AI. Nongovernmental actors cannot solve everything—broad safety nets and education policy are governmental domains—but these actors can pilot solutions, fill gaps, and advocate for communities.

Key actions could include the following:

- **Plan and support workforce transitions.** Businesses that develop or deploy advanced AI could support the employees disrupted by these technologies. This could mean investing in reskilling and upskilling programs, creating apprenticeship or redeployment pathways, and even providing financial cushions (e.g., severance, scholarship funds) for workers displaced by AI. For example, if a company is rolling out an AI system that will automate certain routine jobs, it could concurrently fund training for the employees that used to do those jobs to move into new roles, either in the company or in growing fields externally.

⁷⁵ RAND has published on AI and the social contract. See Benjamin Boudreaux and Beba Cibralic, *Artificial Intelligence and the Social Contract: Foundations for Social and Economic Policy Under Technological Transformation*, RAND Corporation, PE-A3888-1, August 2025.

⁷⁶ For more on legitimacy, see Vivien A. Schmidt, "Democracy and Legitimacy in the European Union Revisited: Input, Output and 'Throughput,'" *Political Studies*, Vol. 61, No. 1, March 2013.

Some tech firms have announced initiatives along these lines, but their impact will be greater if they become standard practice. A lesson from prior automation waves, such as the downturn in manufacturing jobs, is that transition support must be timely, adequately funded, and tailored to local needs. Retraining programs have failed when they are too generic or under-resourced. Research from the past five years on import-competition shocks and displaced-worker policy suggests that regionally concentrated labor-market shocks leave durable local scarring, even when aggregate employment later rebounds. This research also finds that broad, place-blind adjustment measures are insufficient on their own; effective responses require place-based interventions tied to local labor demand, institutional capacity, and complementary investments in regional infrastructure and human capital.⁷⁷

Civil society organizations, such as local nonprofits, worker advocacy groups, and educational institutions could play a role in helping to make these efforts effective. They can reach vulnerable and marginalized workers who might be overlooked and ensure that such programs are accessible (e.g., considering language, location). Without credible transition plans, civil society interviewees warned, communities will resist AI projects, something that has already been seen in localized bans or protests against certain technologies and, relatedly, the creation of new data centers.⁷⁸

- **Use philanthropic capital to build and support AI safety and ethics ecosystem.** There are critical functions that might not have a business model or government support yet, which philanthropic organizations, foundations, charitable organizations, and high-net-worth donors could seed. For instance, funding could support civil society participation in global forums, making sure that diverse voices can attend important meetings on AI governance). Another ripe area is funding training programs or fellowships to grow the community of AI safety auditors, insurers, and technical experts. Philanthropy can also act quickly to support research and interventions on labor and societal impacts—for example, by sponsoring pilot programs for community resilience in areas likely to be hit by automation, which governments can later scale up if shown effective. By filling these gaps, philanthropists can help catalyze and support the safety and societal infrastructure that the market alone might not build—and that government is currently not investing in. One interviewee remarked that in the current AI boom, “funding is going down [for civil society], power is going down,” meaning a few companies are racing ahead and nonprofits struggle to keep up; philanthropic funding injections can level that playing field.
- **Strengthen community-level resilience.** Civil society is often the fabric that holds communities together during times of change. Churches, community centers, local NGOs, and mutual aid networks step in when formal systems are strained. These institutions need to be empowered to help handle AI-related disruptions—whether an influx of unemployed people in a town because of factory automation or a wave of AI-driven scams targeting the elderly in a community. Businesses can support this by providing funding or partnerships to local organizations—for example, by funding a community technology liaison in areas where they deploy AI.

⁷⁷ David Autor, David Dorn, Gordon H. Hanson, Maggie R. Jones, and Bradley Setzler, “Places Versus People: The Ins and Outs of Labor Market Adjustment to Globalization,” National Bureau of Economic Research, NBER Working Paper No. 33424, last updated June 2025; Organisation for Economic Co-operation and Development, “Place-Based Policies for Displaced Workers: Lessons from Past Economic Disruptions,” OECD Local Economic and Employment Development (LEED) Papers, No. 2025/04, June 30, 2025b; Organisation for Economic Co-operation and Development, *Place-Based Policies for the Future*, May 19, 2025a.

⁷⁸ For further exploration of these themes, see Sam Manning’s research. Manning’s research spans the measurement of AI’s labor market and economic impacts and the design of policies to ensure that advanced AI fosters widespread economic prosperity, including work on AI exposure and worker adaptive capacity, international benefit-sharing mechanisms, and the implications of frontier AI model development for labor and inequality (Sam Manning, homepage, undated).

Community-based initiatives can also help track the local effects of AI, such as job losses, changes in service quality, and incidents, and feed information to both companies and policymakers so that emerging harms are caught early. Imagine a city government, local university, and civil society coalition setting up an AI impact task force that meets regularly to review how AI is affecting the city's residents and economy. If the task force identifies, for example, a new AI service that is causing misinformation to spread in a neighborhood, it could notify both the company and public agencies, leading them to intervene. This is analogous to how some communities have environmental monitoring committees near industrial sites. Additionally, community resilience could involve having plans for AI-related crises (which could range from a mass deepfake disinformation incident to an AI-caused critical infrastructure failure). Local emergency preparedness could integrate scenarios involving AI failures or misuse, and here, civil society (similar to volunteer networks and local media) is crucial for effective crisis response.

- **Shape public narratives and inform communication.** In any transformative technological shift, controlling the narrative can influence outcomes as much as controlling the technology. AI has already been subject to hype and fear in media. Misinformation about AI or well-founded concerns left unaddressed can drive public panic or poor decisions in a crisis. A striking quote from one game was, "Whoever controls the communication environment controls the crisis." Civil society can serve as a trusted intermediary to communicate about AI risks and benefits in plain language. For example, respected NGOs or community leaders can translate technical issues (such as what it means if an AI system is "out of control") into terms that make sense for the general public, thus preventing rumors.

During the coronavirus pandemic, partnerships with local community leaders improved trust in public health measures compared with government messaging alone. Similarly, during an AI incident (say, a false alarm of an AI causing some catastrophe), having pre-established communication channels—perhaps a coalition of AI companies, government agencies, and civil society representatives—can ensure that there is consistent, calm messaging.

- **Advocate for inclusive benefits and risk mitigation.** Civil society and businesses could push the principle that those who bear the risks of AI should also share its benefits. If AI dramatically increases productivity or generates huge profits for a few firms, there could be channels, private or public, by which portions of those gains support the broader population. In the context of AI, some ideas on the table include windfall taxes or profit-sharing if an AI is extremely profitable, as well as commitments by AI companies to invest in public goods, including education and digital infrastructure, in proportion to their success.

During the first game, participants explicitly raised questions: "Where is the money going, and shouldn't winners reinvest locally at meaningful scale?" Businesses that are forward-looking might voluntarily set up AI benefit funds that channel a portion of AI-driven gains into community grants, worker training funds, or even universal basic income pilots. For instance, if an AI product replaces local journalism jobs, perhaps the company behind it could fund an independent journalism nonprofit to ensure that community information needs are still met. Or if an AI-driven automation saves a company millions, a slice could go to a community college scholarship program for tech education.

Conditions for success: Efforts under Role 3 will succeed if they are seen as legitimate, collaborative, and sustained. Token or one-off initiatives, such as AI ethics grants that are minor relative to a company's impact, or public relations-heavy commitments that have little follow-through, will be rightly dismissed as window dressing. Partnerships should involve community input—for example, involving worker unions in designing retraining programs or involving local leaders in decisionmaking about how a community fund is spent.

Measurement is also important. Companies could track and publicly report outcomes, such as how many workers they reskilled and how many got new jobs, to prove that these programs have teeth, and civil society

can independently verify those outcomes. Another consideration is avoiding elitism. High-level discussions about reskilling or benefit-sharing must translate to tangible improvements in ordinary people's lives. One interviewee phrased it pointedly: If these programs are "seen as symbolic or elite-driven, they will fail to build trust."

Business and Civil Society Interfaces to Government Action

The three roles outlined in the prior section describe what business and civil society can do on their own authority: manage operational risks, shape market incentives, and support social stability. But these roles do not operate in a vacuum. In practice, some of the most important nongovernmental actions serve a dual purpose; they reduce risk directly *and* they compensate for gaps in government capacity or political will. Where regulation lags, procurement standards and incident-reporting systems do not merely complement public policy—they substitute for it. Where government preparedness is limited, community resilience efforts and workforce transition plans fill needs that public programs have not yet addressed. Over time, however, the aim is not permanent substitution but a working interface: Nongovernmental efforts generate evidence, test approaches, and build institutional capacity that can inform and accelerate smarter regulation when political conditions allow. The following are key ways in which these efforts connect to government action:

- **Evidence generation for smarter regulation:** By instituting such practices as incident reporting, evaluation benchmarks, transparency reports, and audits (Roles 1 and 2), nongovernmental actors generate a wealth of real-world data on AI performance and failures. This evidence is useful for regulators. It allows agencies to move from speculative fears to concrete, measured understanding of AI risks. For example, if an industry-wide incident database reveals that near-loss-of-control events are happening with a certain frequency, regulators could home in on that class of models and mandate specific testing or safety requirements for them.⁷⁹ If insurers' data show that certain mitigations, such as having a human in the loop on critical AI decisions, dramatically reduce losses, that insight could be codified into standards or guidance. In environmental regulation, a precedent is how data from corporate self-monitoring of pollution later helped shape emissions rules.

Already, we see small examples. The AI Incident Database, curated by NGOs, highlights which AI applications (e.g., face recognition, hiring algorithms) are causing the most reported harm; this information can help guide agencies on where to focus oversight. And the AI Risk Management Framework, developed by the National Institute of Standards and Technology, captures important insights from industries actors. Going forward, governments could actively partner in these learning efforts—possibly by providing funding for some of the infrastructure (for example, the National Science Foundation could fund an anonymized incident clearinghouse) and by paying attention to the findings in their rulemaking. This partnership ensures that when governments set binding rules, those rules are evidence-based and targeted to real risks rather than guesswork.

- **Co-regulatory models:** One promising governance approach is co-regulation, in which industry standards and codes of conduct operate in tandem with official regulation. In co-regulation, the government might endorse or recognize an industry-led standard as meeting regulatory requirements, essentially outsourcing some of the details to those closer to the technology while keeping a legal backstop. The EU's General-Purpose AI Code of Practice illustrates this approach: Its chapters on transparency and copyright apply broadly to general-purpose AI providers, while its safety-and-security chapter targets the most advanced systemic risk models with concrete risk-management commitments, creating an

⁷⁹ AI Incident Database, homepage, undated-b.

interim compliance regime in advance of the AI Act's full enforcement. In the United States, state-level developments, such as California's Senate Bill 53 and New York's RAISE Act,⁸⁰ both of which require frontier developers to publish safety frameworks and report critical incidents, are creating disclosure obligations that nongovernmental safety practices can directly feed into.

- **Regulatory sandboxes and innovation-enabling partnerships:** Governments can also directly collaborate with companies and civil society through regulatory sandboxes, pilot programs, or memorandums of understanding that allow controlled experimentation with new AI technologies. Early memorandums of understanding between leading labs and safety institutes demonstrate that structured regulator-lab access is feasible. It is worth exploring how these kinds of partnerships can be expanded, both in size and scale, between government and labs, as well as beyond the labs. For example, deployers of AI models in critical domains, such as finance and health care, could work with government to pilot safety testing of AI models in specific use-cases.

A health care regulator might set up a sandbox in which a hospital and an AI company can deploy a novel diagnostic AI on a limited trial basis with certain guardrails—for example, extra monitoring, a safety review board oversight, and a requirement to share outcome data—even if formal approval processes are not yet fully defined for that kind of AI. In exchange for this regulatory flexibility, the company agrees to transparently share results and lessons learned with the regulator and perhaps the public. This way, rules can be co-developed based on actual pilot experience rather than abstract scenarios.

- **Addressing systemic and international challenges:** There are areas in which even the best nongovernmental efforts will not suffice and direct government action is needed, but the groundwork laid by those efforts helps inform and motivate that action. One such area is systemic risk mitigation. If a few AI models become so powerful or ubiquitous that their failure would be catastrophic (such as in an AGI scenario), governments might need to impose binding constraints, such as requiring licenses for certain safety levels before deployment. The data and norms from industry can guide what those constraints should be. Another area is the concentration of resources. For instance, if cutting-edge AI is in the hands of only two giant firms with immense compute power, government might consider enforcing antitrust regulation or opening up access.

The relationship runs in both directions. Public policy can make the roles described here more effective, providing safe harbors for incident reporting and the recognition of voluntary certifications or cofunding transition programs to lower barriers to adoption. But some of the actions that we recommend exist to fill gaps that government has left open. If governments were to establish mandatory disclosure requirements, fund workforce transitions at scale, or set binding safety standards with adequate enforcement, the case for nongovernmental actors to take on those functions would diminish accordingly. That would be a success, not a loss. The purpose of this framework is not to define a permanent division of labor but to identify what nongovernmental actors can and should do and to generate the evidence and institutional capacity that better public policy may eventually require.

⁸⁰ California Senate Bill 53, Artificial Intelligence Models: Large Developers, September 29, 2025; New York State Senate Bill S6953A, An Act to Amend the General Business Law, in Relation to the Training and Use of Artificial Intelligence Frontier Models, December 19, 2025.

Conclusion

The evidence gathered in this report points to a consistent finding: Managing transformative AI requires action from business and civil society, not only from governments. The risks are technical and social, fast-moving and unevenly distributed, and they emerge where systems are designed, deployed, and used—thus, well within the reach of nongovernmental actors. The framework that we propose—managing technical and operational risks, shaping incentives through markets and networks, and supporting social stability and public trust—is intended to give actors across business and civil society a practical, structured way to identify what they can do now under existing legal and market conditions, without waiting for regulatory mandates to arrive.

Progress in each role depends on building governance infrastructure that makes good practice routine rather than exceptional. That could include board-level oversight and documented safety cases inside firms; procurement criteria, assurance signals, and investor diligence across markets; and workforce transition planning and community partnerships in the social domain. These are not novel ideas in isolation; elements of these ideas are already being piloted across the ecosystem. However, they remain fragmented and ad hoc. The contribution of such a framework as this one is to generalize those practices, specify minimum baselines that are actionable across actor types, and identify where coordination among actors is needed to make individual efforts stick.

We especially hope that actors who have not yet engaged directly with transformative AI risk will find in this framework a concrete entry point. Whether or not regulation catches up, the capacity that these actors build now will determine how well societies manage what is coming.

Semistructured Interviews

We conducted 15 semistructured interviews with stakeholders across sectors in mid-2025. Participants were selected via purposive sampling to cover a variety of perspectives relevant to AI governance outside government. Participants included senior managers at tech companies (e.g., responsible AI leads, product safety engineers), executives in sectors that are adopting AI (e.g., health care, finance) who deal with AI risk, leaders of AI-focused civil society organizations and think tanks, academics who focus on AI ethics, and former government officials who are now in private sector roles (e.g., a former cybersecurity official in a tech company). Participants were recruited via direct outreach and professional referrals. Inclusion criteria emphasized senior decisionmakers or SMEs with responsibilities related to AI development, deployment, or oversight across business, civil society, and (for triangulation) selected government roles. We summarize the content of these interviews in Table A.1.

Interviews lasted between 30 and 60 minutes, were not recorded, and were analyzed for key themes and insights. To encourage candor, we offered anonymity for specific quotes (hence, we cite quotes without attribution to individual names or organizations). Key emergent themes—such as the *dual horizon* of near-term versus existential AI concerns, the emphasis on demand-side incentives, and warnings about labor displacement and fragmentation of effort—are reflected in the report.

The interview protocol covered such topics as perceived AI risks in the near versus long term, existing practices for managing AI risk in organizations, views on what industry or civil society could do beyond the status quo, dependencies on government action, and concerns about feasibility or unintended consequences of nongovernmental interventions.

After introductory questions on the interviewee’s background, we asked, “(1) In your view, what are the most pressing risks arising from advanced AI in the next five years? Ten-plus years?” followed by “(2) How

TABLE A.1
Semistructured Interview Participants and Focus Areas

Sector	Total Participants	Focus Areas (with counts and brief descriptions)
Business	5	- Product safety (3): corporate AI governance, model assurance, and compliance practices - Finance (1): risk modeling and AI oversight in financial institutions - National security (1): applications of AI in defense and intelligence contexts
Civil society	7	- Higher education (2): academic governance, ethics, and research policy - Advocacy (1): nonprofit initiatives on AI accountability - Consulting (1): advisory roles on AI governance and risk management - Health care (1): responsible AI in clinical settings - International organizations (1): multilateral coordination on AI standards - AI safety (1): independent safety research and public communication
Government	3	- Cybersecurity (1): resilience and threat mitigation in digital infrastructure - Local government (1): municipal-level AI adoption and oversight - Export controls (1): regulation of advanced AI models and dual-use technologies
Total	15	

adequately equipped are existing institutions (in your sector or community) to manage those risks?” We then explored nonstate roles: “(3) What actions or policies by companies could significantly reduce AI-related risks? What about actions by professional associations, nonprofits, or other civil [society] groups?” To probe incentives, we asked, “(4) Are there market or social pressures that could encourage safer AI development (e.g., from investors, customers, insurance)?” On feasibility, we asked, “(5) What barriers would a company or nonprofit face if it tried to implement the ideas we’ve discussed?” And we closed with scenario prompts, such as “(6) Imagine a serious AI incident has occurred. How do you think industry and civil society would respond, and what would you wish had been in place?”

The study followed institutional human-subjects protocols; participants provided informed consent. Data were stored on encrypted drives with access restricted to the research team. We minimized the collection of personally identifiable information; used sector, region, and domain labels in place of names; and will retain de-identified notes for three months before secure deletion.

The sample is weighted toward U.S. perspectives and product-safety roles; several stakeholder types (e.g., labor, insurers, cloud and infrastructure providers, standards bodies, Global South) are underrepresented. A lack of audio recordings may limit verbatim precision. Findings are qualitative and not statistically generalizable.

Day After AGI

A separate after-action report details research findings from the games.¹ Because observations from the games were considered in the development of the framework, in the following paragraphs, we briefly describe the structure of the gaming platform. More-detailed information is shared in the other report.

To understand how the United States should respond to and prepare for potential artificial intelligence (AI) and artificial general intelligence (AGI) developments in the future, the RAND Geopolitics of AGI Initiative is conducting a series of Day After AGI exercises using the RAND Infinite Potential platform. These exercises support the initiative's overall goal of understanding how the United States can respond to the technically credible but uncertain potential for AGI to be developed. Specifically, these exercises have the objective of understanding how the United States might wish to respond to sudden changes brought about by AI and, in particular, what judgments participants seek to make, what capabilities they believe they need to have on hand in the scenario presented to them, and what playbooks they believe they need to develop to guide their response.

Infinite Potential Day After AGI exercises are two-hour tabletop exercises patterned after the Day After format, with ten to 16 participants led by a facilitator whose job is to elicit responses and discussion from the group on one of several scenarios regarding AGI. Participants are selected based on their career, institutional experience, and knowledge.

Most exercises simulate a National Security Council Principals Committee (PC) convening to recommend a U.S. government response to developments in frontier AI. In Business and Civil Society games, the exercise simulates the convening of a fictional Business and Civil Society Council that is convened at the pleasure of the U.S. President in the wake of a crisis. In each exercise, participants are presented with a scenario that represents both (1) an acute crisis for U.S. national or economic security and (2) a signpost on a path to a transformative AI future. With facilitation by a simulated National Security Advisor, participants role-play as business and civil society leaders diagnosing the implications of the scenario for U.S. national security, discussing courses of action and the roles business and civil society institutions might take to mitigate risks, and recommending a path forward for the President of the United States.

In the exercise, participants face a predeveloped scenario over two turns. In turn 1, participants are asked to respond to intelligence reports and openly available information about an unfolding crisis involving AGI and to discuss the immediate policy ramifications. Teams are briefed by a subject-matter expert on the crisis and can ask clarifying questions about the events. The facilitator then leads the group through a discussion about policy objectives, options for action, and ramifications of potential decisions. All discussions are captured by notetakers. In turn 2, the scenario progresses, and the crisis takes a turn, usually for the worse, but sometimes just in an unexpected vector. Teams are again briefed by an expert on the accelerated crisis and can ask questions and dig into the policy issues and options for action. At the end of turn 2, the facilitator prompts participants to retrospectively evaluate their initial responses and strategies. In some iterations of

¹ Girven, Cibralic, and Smith, 2026.

the exercise, the teams are asked to backcast to consider what could be done prior to the events unfolding to prevent them from happening or to mitigate the effects should they be inevitable.

Gameplay is set in a hypothetical near future of 12 to 18 months from the date of the exercise. This relatively quick time frame was selected because AI's rapid pace of improvement creates a desire to explore what a potential AI crisis could look like in the short term rather than on a longer timeline. Participants are instructed to assume that the technological and geopolitical landscape will not have changed significantly, with identified vulnerabilities remaining comparable with those that exist today. The overall tenor of U.S. national politics and the foreign engagement environment is like that of today, with specific changes indicated in the lead-in scenario. This means that, unless otherwise specified in the scenario, the realities of the current year persist in the exercise.

Abbreviations

AGI	artificial general intelligence
AI	artificial intelligence
ESG	Environmental, Social, and Governance
EU	European Union
IEC	International Electrotechnical Commission
ISO	International Organization for Standardization
LLM	large language model
NGO	nongovernmental organization
NIST	National Institute of Standards and Technology
R&D	research and development
SME	subject-matter expert

References

- Adan, Sumaya Nur, Joanna Wiaterek, Varun Sen Bahl, Ima Bello, Camila Beltran, Tobias Dierks, Luise Eder, Liam Epstein, Abra Ganz, Natalie Kiilu, et al., *AI Benefit-Sharing Framework: Balancing Access and Safety*, Oxford Martin AI Governance Initiative, Oxford Martin School, University of Oxford, December 1, 2025.
- AI Incident Database, “About,” webpage, Responsible AI Collaborative, undated-a. As of April 20, 2026: <https://incidentdatabase.ai/about/>
- AI Incident Database, homepage, undated-b. As of May 14, 2026: <https://incidentdatabase.ai/>
- Alaga, Jide, and Jonas Schuett, “Coordinated Pausing: An Evaluation-Based Coordination Scheme for Frontier AI Developers,” arXiv, arXiv:2310.00374, September 30, 2023.
- Amanpour & Company, “Geoffrey Hinton Warns of the ‘Existential Threat’ of AI,” video, May 9, 2023. As of April 17, 2026: <https://www.pbs.org/video/godfather-of-ai-warns-of-the-existential-threat-of-ai-ljl1lc/>
- Amazon News, “We Are Implementing a One-Year Moratorium on Police Use of Rekognition,” news release, June 10, 2020.
- Anthony, Aubra, Lakshmee Sharma, and Elina Noor, *Advancing a More Global Agenda for Trustworthy Artificial Intelligence*, Carnegie Endowment for International Peace, April 30, 2024.
- Anthropic, “Anthropic’s Responsible Scaling Policy,” webpage, version 2.2, effective May 14, 2025. As of April 17, 2026: <https://www.anthropic.com/responsible-scaling-policy>
- “Artificial Intelligence (AI) Legislation Tracker 2026: All 50 States,” webpage, MultiState.ai, undated. As of May 14, 2026: <https://www.multistate.ai/artificial-intelligence-ai-legislation>
- Artificial Intelligence Underwriting Company, homepage, undated. As of April 17, 2026: <https://aiuc.com/>
- Autor, David, David Dorn, Gordon H. Hanson, Maggie R. Jones, and Bradley Setzler, “Places Versus People: The Ins and Outs of Labor Market Adjustment to Globalization,” National Bureau of Economic Research, NBER Working Paper No. 33424, last updated June 2025.
- Balleisen, Edward J., and Marc A. Eisner, “The Promise and Pitfalls of Co-Regulation: How Governments Can Draw on Private Governance for Public Purpose,” in David Moss and John Cisternino, eds., *New Perspectives on Regulation*, The Tobin Project, 2009.
- Bengio, Yoshua, Stephen Clare, Carina Prunkl, Maksym Andriushchenko, Ben Bucknell, Malcolm Murray, Shalaleh Rismani, Conor McGlynn, Nestor Maslej, Philip Fox, et al., *International AI Safety Report 2026*, February 2026.
- Berg, Florian, Julian F. Kölbel, and Roberto Rigobon, “Aggregate Confusion: The Divergence of ESG Ratings,” *Review of Finance*, Vol. 26, No. 6, November 2022.
- Boudreaux, Benjamin, and Beba Cibralic, *Artificial Intelligence and the Social Contract: Foundations for Social and Economic Policy Under Technological Transformation*, RAND Corporation, PE-A3888-1, August 2025. As of April 17, 2026: <https://www.rand.org/pubs/perspectives/PEA3888-1.html>
- Brundage, Miles, Noemi Dreksler, Aidan Homewood, Sean McGregor, Patricia Paskov, Conrad Stosz, Girish Sastry, A. Feder Cooper, George Balston, Steven Adler, et al., *Frontier AI Auditing: Toward Rigorous Third-Party Assessment of Safety and Security Practices at Leading AI Companies*, January 2026.
- California Senate Bill 53, Artificial Intelligence Models: Large Developers, September 29, 2025.
- Campaign to Stop Killer Robots, “Urgent Action Needed to Ban Fully Autonomous Weapons: Non-Governmental Organizations Convene to Launch Campaign to Stop Killer Robots,” press release, April 23, 2013.
- Chambers, Simone, and Jeffrey Kopstein, “Bad Civil Society,” *Political Theory*, Vol. 29, No. 6, December 2001.

- Chan, Alan, Kevin Wei, Sihao Huang, Nitarshan Rajkumar, Elija Perrier, Gillian K. Hadfield, and Markus Anderljung, "Infrastructure for AI Agents," arXiv, arXiv:2501.10114, version 2, May 16, 2025.
- Cibralic, Beba, Miranda Bogen, Kevin Bankston, and Ruchika Joshi, *Principled Practice: A Playbook for Operationalizing Responsible AI*, Center for Democracy & Technology, AI Governance Lab, September 17, 2025.
- Cloud Security Alliance, "Cloud Security Alliance Launches STAR for AI, Establishing the Global Framework for Responsible and Auditable Artificial Intelligence," press release, October 23, 2025.
- Croxtton, John, David Robusto, Satya Thallam, and Doug Calidas, "Message Incoming: Establish an AI Incident Reporting System," Federation of American Scientists, September 26, 2024.
- CVE Program, "Overview: About the CVE Program," webpage, undated. As of April 17, 2026: <https://www.cve.org/about/overview>
- Dafoe, Allan, *AI Governance: A Research Agenda*, Centre for the Governance of AI, Future of Humanity Institute, University of Oxford, August 27, 2018.
- Dastin, Jeffrey, and Aditya Soni, "Sam Altman to Return as OpenAI CEO After His Tumultuous Ouster," Reuters, November 22, 2023.
- Delmas, Magali A., and Vanessa C. Burbano, "The Drivers of Greenwashing," *California Management Review*, Vol. 54, No. 1, October 2011.
- Dennis, Claire, Sam Manning, Stephen Clare, Boxi Wu, Jake Okechukwu Effoduh, Chinasa T. Okolo, Lennart Heim, and Katya Klinova, *Options and Motivations for International AI Benefit Sharing*, Centre for the Governance of AI, Future of Humanity Institute, University of Oxford, February 6, 2025.
- Department for Science, Innovation and Technology, "Frontier AI Safety Commitments, AI Seoul Summit 2024," United Kingdom government guidance, last updated February 7, 2025.
- Department of Defense Standard Practice MIL-STD-882E, *System Safety*, incorporating change 1, U.S. Department of Defense, September 27, 2023.
- Dreksler, Noemi, Harry Law, Chloe Ahn, Daniel S. Schiff, Kaylyn Jackson Schiff, and Zachary Peskowitz, *What Does the Public Think About AI? An Overview of the Public's Attitudes Towards AI and a Resource for Future Research*, Centre for the Governance of AI, Future of Humanity Institute, University of Oxford, January 22, 2025.
- Dryzek, John S., "Global Civil Society: The Progress of Post-Westphalian Politics," *Annual Review of Political Science*, Vol. 15, 2012.
- Erman, Eva, and Markus Furendal, "The Democratic Role of Non-State Actors in the Global Governance of Artificial Intelligence," *Critical Review of International Social and Political Philosophy*, online ahead of print, 2025.
- European Commission, "General-Purpose AI Code of Practice Now Available," news release, July 10, 2025a.
- European Commission, *The General-Purpose AI Code of Practice*, July 10, 2025b. As of May 7, 2026: <https://digital-strategy.ec.europa.eu/en/policies/contents-code-gpai>
- European Commission, "Questions and Answers on the Code of Practice for General-Purpose AI," July 11, 2025c.
- European Commission, "Drawing-Up a General-Purpose AI Code of Practice," last updated March 23, 2026.
- Executive Order 14110, "Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence," Executive Office of the President, October 30, 2023.
- Executive Order 14179, "Removing Barriers to American Leadership in Artificial Intelligence," Executive Office of the President, January 23, 2025.
- Finck, Michèle, "Digital Co-Regulation: Designing a Supranational Legal Framework for the Platform Economy," *European Law Review*, Vol. 43, No. 1, 2018.
- Frontier Model Forum, "Progress Update: Advancing Frontier AI Safety in 2024 and Beyond," August 29, 2024.
- Future of Life Institute, "Pause Giant AI Experiments: An Open Letter," March 29, 2023.

- Galvagna, Christine, *Inclusive AI Governance: Civil Society Participation in Standards Development*, Ada Lovelace Institute, March 30, 2023.
- Girven, Richard S., Beba Cibralic, and Gregory Smith, *Infinite Potential—Insights from the Business and Civil Society Scenario: After-Action Report from a Sequence of Day After Artificial General Intelligence Exercises*, RAND Corporation, RR-A4767-1, 2026. As of April 20, 2026: https://www.rand.org/pubs/research_reports/RRA4767-1.html
- Global Partnership on Artificial Intelligence, homepage, Organisation for Economic Co-operation and Development, undated. As of May 7, 2026: <https://gpai.ai/>
- Google DeepMind, “Frontier Safety Framework,” version 3.1, April 17, 2026.
- Governor Gavin Newsom, “Governor Newsom Signs SB 53, Advancing California’s World-Leading Artificial Intelligence Industry,” news release, September 29, 2025.
- Governor Kathy Hochul, “Governor Hochul Signs Nation-Leading Legislation to Require AI Frameworks for AI Frontier Models,” news release, December 19, 2025.
- Greene, Jay, “Microsoft Won’t Sell Police Its Facial-Recognition Technology,” *Washington Post*, June 11, 2020.
- Gudgel, John E., “Insurance and the Public–Private Management of Risk at US Commercial Nuclear Power Plants,” *Risk Management and Insurance Review*, Vol. 26, No. 4, Winter 2023.
- Harvard Law Review Association, “Amoral Drift in AI Corporate Governance,” *Harvard Law Review*, Vol. 138, No. 6, April 2025.
- Hendrycks, Dan, Dawn Song, Christian Szegedy, Honglak Lee, Yarin Gal, Erik Brynjolfsson, Sharon Li, Andy Zou, Lionel Levine, Bo Han, et al., “A Definition of AGI,” arXiv, arXiv:2510.18212, version 2, revised October 23, 2025.
- Homewood, Aidan, Sophie Williams, Noemi Dreksler, John Lidiard, Malcolm Murray, Lennart Heim, Marta Ziosi, Seán Ó hÉigeartaigh, Michael Chen, Kevin Wei, et al., “Third-Party Compliance Reviews for Frontier AI Safety Frameworks,” arXiv, arXiv:2505.01643, version 1, May 3, 2025.
- IEEE—See Institute of Electrical and Electronics Engineers.
- Institute of Electrical and Electronics Engineers, *Ethically Aligned Design: A Vision for Prioritizing Human Well-Being with Autonomous and Intelligent Systems*, April 2019.
- Institute of Internal Auditors, *The Three Lines of Defense in Effective Risk Management and Control*, January 2013.
- International Organization for Standardization and International Electrotechnical Commission, “Information Technology—Artificial Intelligence—Management System,” ISO/IEC 42001:2023, 2023.
- Kerry, Cameron F., Joshua P. Meltzer, Andrea Renda, and Andrew Wyckoff, “A Network Architecture for Global AI Policy,” Brookings Institution, February 10, 2025.
- Kilian, Robert, Linda Jäck, and Dominik Ebel, “European AI Standards—Technical Standardisation and Implementation Challenges Under the EU AI Act,” *European Journal of Risk Regulation*, Vol. 16, No. 3, September 2025.
- Kolt, Noam, Nicholas Caputo, Jack Boeglin, Cullen O’Keefe, Rishi Bommasani, Stephen Casper, Mariano-Florentino Cuéllar, Noah Feldman, Iason Gabriel, Gillian K. Hadfield, et al., *Legal Alignment for Safe and Ethical AI*, Oxford Martin School, AI Governance Initiative, University of Oxford, January 15, 2026.
- Lall, Ranjit, “Making Global Governance Accountable: Civil Society, States, and the Politics of Reform,” *American Journal of Political Science*, Vol. 69, No. 1, January 2025.
- Longpre, Shayne, Sayash Kapoor, Kevin Klyman, Ashwin Ramaswami, Rishi Bommasani, Borhane Billi-Hamelin, Yangsibo Huang, Aviya Skowron, Zheng-Xin Yong, Suhas Kotha, et al., “Position: A Safe Harbor for AI Evaluation and Red Teaming,” *Proceedings of the 41st International Conference on Machine Learning*, July 2024.
- Lyon, Thomas P., and A. Wren Montgomery, “The Means and End of Greenwash,” *Organization & Environment*, Vol. 28, No. 2, June 2015.

Manning, Sam, homepage, undated. As of April 20, 2026:

<https://www.sam-manning.com/>

METR, "Frontier AI Safety Policies," webpage, undated. As of April 20, 2026:

<https://metr.org/fsp>

MITRE, "CVE Program Celebrates 25 Years of Impact in Cybersecurity," news release, October 22, 2024.

Mökander, Jakob, Jonas Schuett, Hannah Rose Kirk, and Luciano Floridi, "Auditing Large Language Models: A Three-Layered Approach," *AI and Ethics*, Vol. 4, No. 4, November 2024.

Montgomery, A. W., Thomas P. Lyon, and Jasper Barg, "No End in Sight? A Greenwash Review and Research Agenda," *Organization & Environment*, Vol. 37, No. 2, June 2024.

Morrone, Megan, "Amodei on AI: 'There's a 25% Chance That Things Go Really, Really Badly,'" *Axios*, September 17, 2025.

National Aeronautics and Space Administration, "Aviation Safety Reporting System (ASRS): Program Briefing," webpage, undated. As of April 20, 2026:

<https://asrs.arc.nasa.gov/overview/summary.html>

National Aeronautics and Space Administration, *Reliability Program Requirements for Aeronautical and Space System Contractors*, NHB 5300.4(1A-1), January 1987.

National Institute of Standards and Technology, "Center for AI Standards and Innovation (CAISI)," webpage, undated. As of May 20, 2026:

<https://www.nist.gov/caisi>

National Institute of Standards and Technology, "Fact Sheet: U.S. Department of Commerce & U.S. Department of State Launch the International Network of AI Safety Institutes at Inaugural Convening in San Francisco," November 20, 2024.

New York State Senate Bill S6953A, An Act to Amend the General Business Law, in Relation to the Training and Use of Artificial Intelligence Frontier Models, December 19, 2025.

Office for Artificial Intelligence, *Guidelines for AI Procurement*, United Kingdom government guidance, June 8, 2020.

OpenAI, *Preparedness Framework*, version 2, last updated April 15, 2025.

Organisation for Economic Co-operation and Development, "Recommendation of the Council on Artificial Intelligence," OECD/LEGAL/0449, amended May 3, 2024.

Organisation for Economic Co-Operation and Development, *Place-Based Policies for the Future*, May 19, 2025a.

Organisation for Economic Co-Operation and Development, "Place-Based Policies for Displaced Workers: Lessons from Past Economic Disruptions," OECD Local Economic and Employment Development (LEED) Papers, No. 2025/04, June 30, 2025b.

Partnership on AI, homepage, undated. As of May 7, 2026:

<https://partnershiponai.org/>

Perrigo, Billy, and Tharin Pillay, "'Time Is Running Out': New Open Letter Calls for Ban on Superintelligent AI Development," *Time*, October 21, 2025.

Png, Marie-Therese, "At the Tensions of South and North: Critical Roles of Global South Stakeholders in AI Governance," *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency*, Association for Computing Machinery, 2022.

Rao, Anita, Drew Keller, Neha Kalra, Ryan Steed, Kweku Kwegyir-Aggrey, Kevin Klyman, Diane Staheli, and Stevie Bergman, *Challenges to the Monitoring of Deployed AI Systems*, Center for AI Standards and Innovation, National Institute of Standards and Technology, NIST AI 800-4, updated March 18, 2026.

Reavis, Jim, "Announcing the AI Controls Matrix and ISO/IEC 42001 Mapping—and the Roadmap to STAR for AI 42001," Cloud Security Alliance, August 20, 2025.

Responsible Artificial Intelligence Institute, homepage, undated. As of April 20, 2026:

<https://www.responsible.ai/>

- Reuel, Anka, Ben Bucknall, Stephen Casper, Tim Fist, Lisa Soder, Onni Aarne, Lewis Hammond, Lujain Ibrahim, Alan Chan, Peter Wills, et al., “Open Problems in Technical AI Governance,” arXiv, arXiv:2407.14981, version 2, April 16, 2025.
- Risk & Insurance, “Cyber Insurance Provides Both Carrot and Stick for Cybersecurity,” July 11, 2024.
- Schmidt, Vivien A., “Democracy and Legitimacy in the European Union Revisited: Input, Output and ‘Throughput,’” *Political Studies*, Vol. 61, No. 1, March 2013.
- Schuett, Jonas, “Three Lines of Defense Against Risks from AI,” *AI and Society*, Vol. 40, No. 2, February 2025.
- Schuett, Jonas, Ann-Katrin Reuel, and Alexis Carlier, “How to Design an AI Ethics Board,” *AI and Ethics*, Vol. 5, No. 2, April 2025.
- Schultz, Mario D., Ludovico Giacomo Conti, and Peter Seele, “Digital Ethicswashing: A Systematic Review and a Process-Perception-Outcome Framework,” *AI and Ethics*, Vol. 5, No. 2, April 2025.
- Schwarcz, Daniel, and Josephine Wolff, “The Limits of Regulating AI Safety Through Liability and Insurance: Lessons from Cybersecurity,” *Fordham Law Review*, 2026.
- Seetharaman, Deepa, Stephen Nellis, and Deborah Mary Sophia, “Microsoft, OpenAI Reach Deal Removing Fundraising Constraints for ChatGPT Maker,” Reuters, October 28, 2025.
- Shackelford, Scott J., and Rachel Dockery, “Governing AI,” *Cornell Journal of Law & Public Policy*, Vol. 30, No. 2, Winter 2020.
- Simpson, Morgan, Alejandro Ortega, and Robert Trager, “Voluntary Industry Initiatives in Frontier AI Governance: Lessons from Aviation and Nuclear Power,” AI Governance Initiative, Oxford Martin School, University of Oxford, January 20, 2025.
- Skey, Michael, “Sportswashing: Media Headline or Analytic Concept?” *International Review for the Sociology of Sport*, Vol. 58, No. 5, August 2023.
- Soares, Nate, Benja Fallenstein, Stuart Armstrong, and Eliezer Yudkowsky, “Corrigibility,” *AAAI Workshops: Workshops at the Twenty-Ninth AAAI Conference on Artificial Intelligence*, 2015.
- Stosz, Conrad, Karson Elmgren, Charles Foster, George Balston, Seth Donoughe, Samira Nedungadi, Michael Chen, Jasper Götting, Patricia Paskov, Sayash Kapoor, et al., *AEF-1: Minimum Operating Conditions for Independent Third Party AI Evaluations*, version 1, AI Evaluator Forum, undated.
- Stout, Lynn A., *The Shareholder Value Myth: How Putting Shareholders First Harms Investors, Corporations, and the Public*, Berrett-Koehler Publishers, 2012.
- Taylor, Bret, “Built to Benefit Everyone,” OpenAI, October 28, 2025.
- Trout, Cristian, “Liability and Insurance for Catastrophic Losses: The Nuclear Power Precedent and Lessons for AI,” *Generative AI and Law Workshop at the International Conference on Machine Learning*, 2024.
- Trout, Cristian, “Healthy Insurance Markets Will Be Critical for AI Governance,” *Lawfare* blog, December 17, 2025.
- U.S.–Canada Power System Outage Task Force, *Final Report on the August 14, 2003 Blackout in the United States and Canada: Causes and Recommendations*, April 2004.
- U.S. Code, Title 15, Section 1, Trusts, Etc., in Restraint of Trade Illegal; Penalty (Sherman Act).
- Wareham, Mary, *Stopping Killer Robots: Country Positions on Banning Fully Autonomous Weapons and Retaining Human Control*, Human Rights Watch, August 10, 2020.
- Wei, Kevin, and Lennart Heim, “Designing Incident Reporting Systems for Harms from General-Purpose AI,” arXiv, arXiv:2511.05914, version 1, November 8, 2025.
- World Economic Forum, “AI Global Alliance,” webpage, undated. As of April 20, 2026: <https://initiatives.weforum.org/ai-global-alliance/home>
- World Economic Forum, *Unlocking Public Sector AI: AI Procurement in a Box: AI Government Procurement Guidelines*, toolkit, 2020.

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